AQUABOTS Aeropak

Critical Design Review

Team 26061

THE UNIVERSITY OF ARIZONA

Overview

1.0 Introduction

2.0 System Requirements

3.0 Point Design

4.0 Performance Analyses and Expectations

5.0 Verification

6.0 Risk Analysis

7.0 Project Status and Plan

8.0 CDE 1

9.0 CDE 2

10.0 Cost Analysis and Deliverables

11.0 Summary

Introduction

Our Team

- **Jamie Ledbetter, Biosystems Engineering – Team Lead**
- **Robert Wilkie, Aerospace Engineering – Procurement Lead**
- **Alex Imrich, Aerospace Engineering**
- **John Lopez, Electrical Engineering**
- **Mayra Mendoza, Optical Engineering**
- **Kama Henriques, Mechanical Engineering**

Project Summary

What makes AQUABOT different?

- Rapid-response
- Short-term, targeted missions
- Swarm control capabilities
- Scalable

Key Features:

- Mounted and fully functional sensor/camera suite
- Further developed communications
- Easy Integration with other water vessels



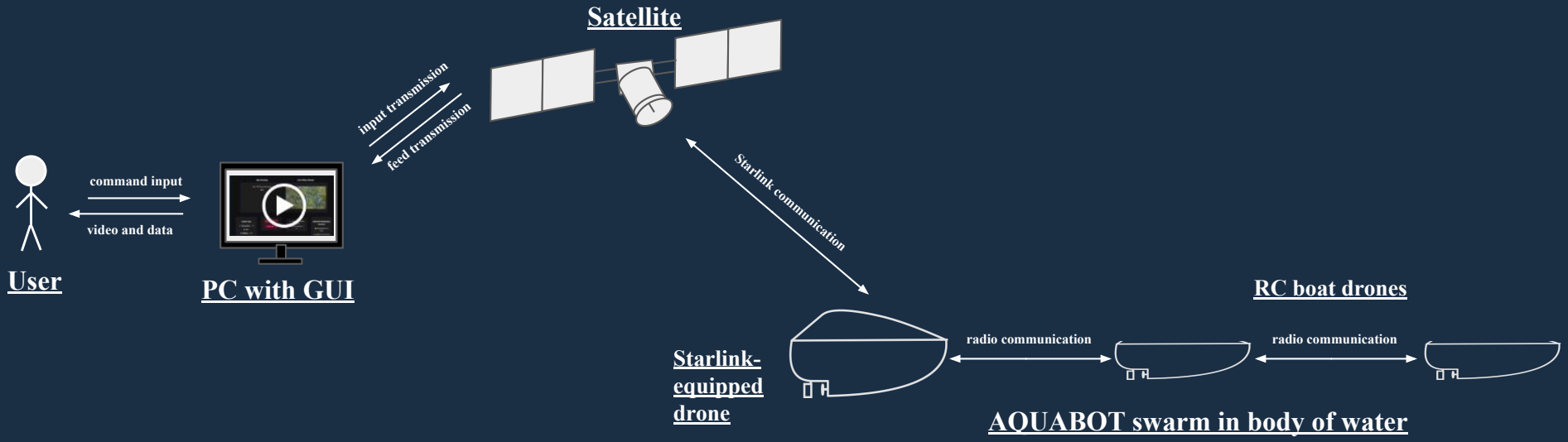
Image of AQUABOT Hull

Sponsors: Dr. Marvin Slepian, Eddy Stocker

Project Summary: OV-1

Application:

- Deployable in lakes and the ocean
- Remote data acquisition and swarm command



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PDR Action Items

Action Items

1. Add a system requirement for drone to drone communication
2. Add a system requirement for swarm control
3. Update SBD
4. Generate a weight analysis
5. Define/resolve deployment system approach with your sponsor

All Action Items Closed

Action Taken

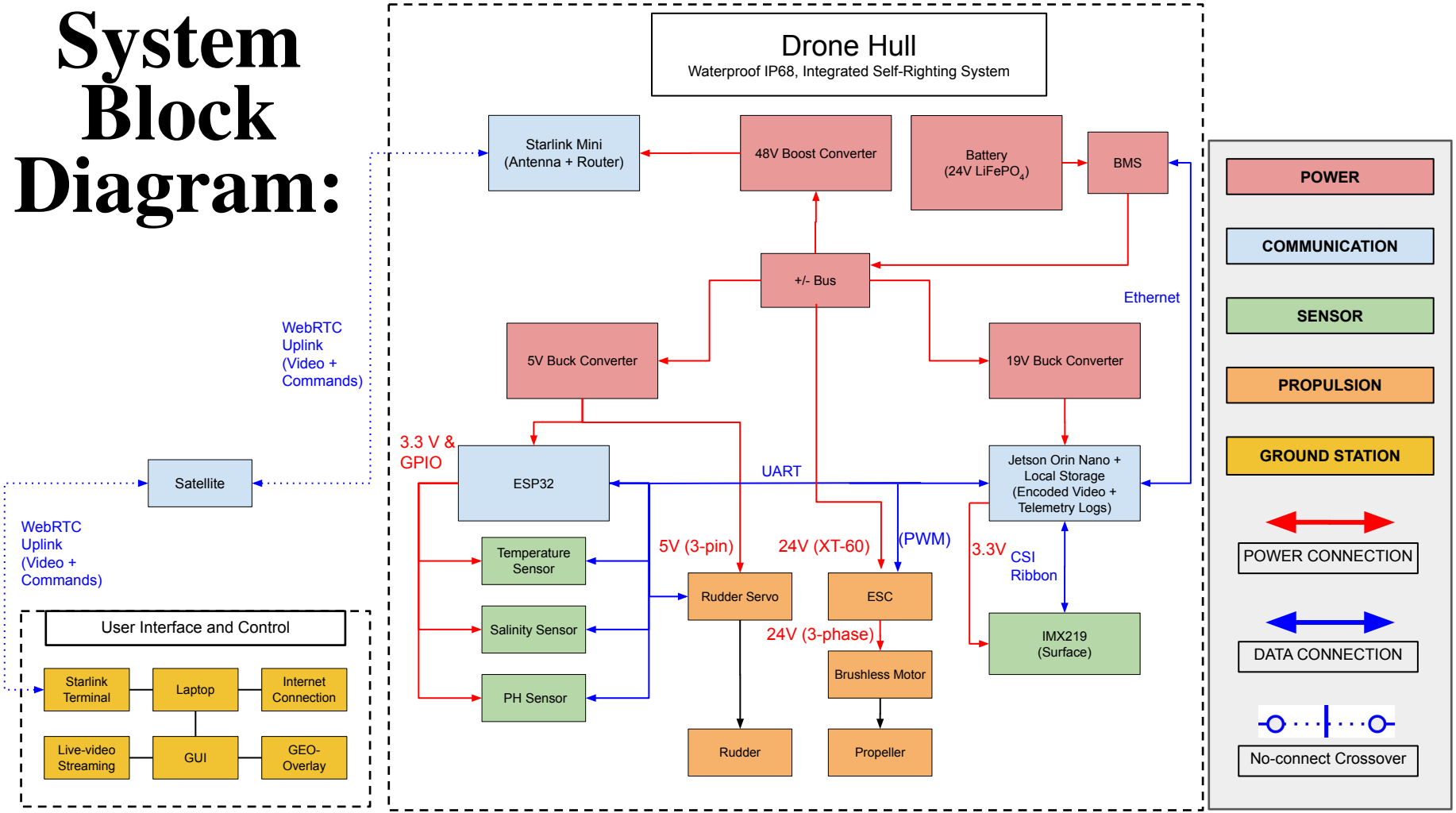
1. Created Requirement 5.4.2 for inter-drone communication up to 100m
2. Created Requirement 5.4.6 for coordinated drone swarm behavior with a leader-follower model at < 1 s latency
3. Updates made to SBD:
   a. Modification to reflect finalized motor and prop layout
   b. Removal of in-drone E-stop and Master Switch
4. Weight Analysis conducted to find total drone mass of ~37 kg
5. Team negotiation and discussion with sponsors resulted in a change of focus from deployment system to swarm demonstration

Major Post-PDR Changes

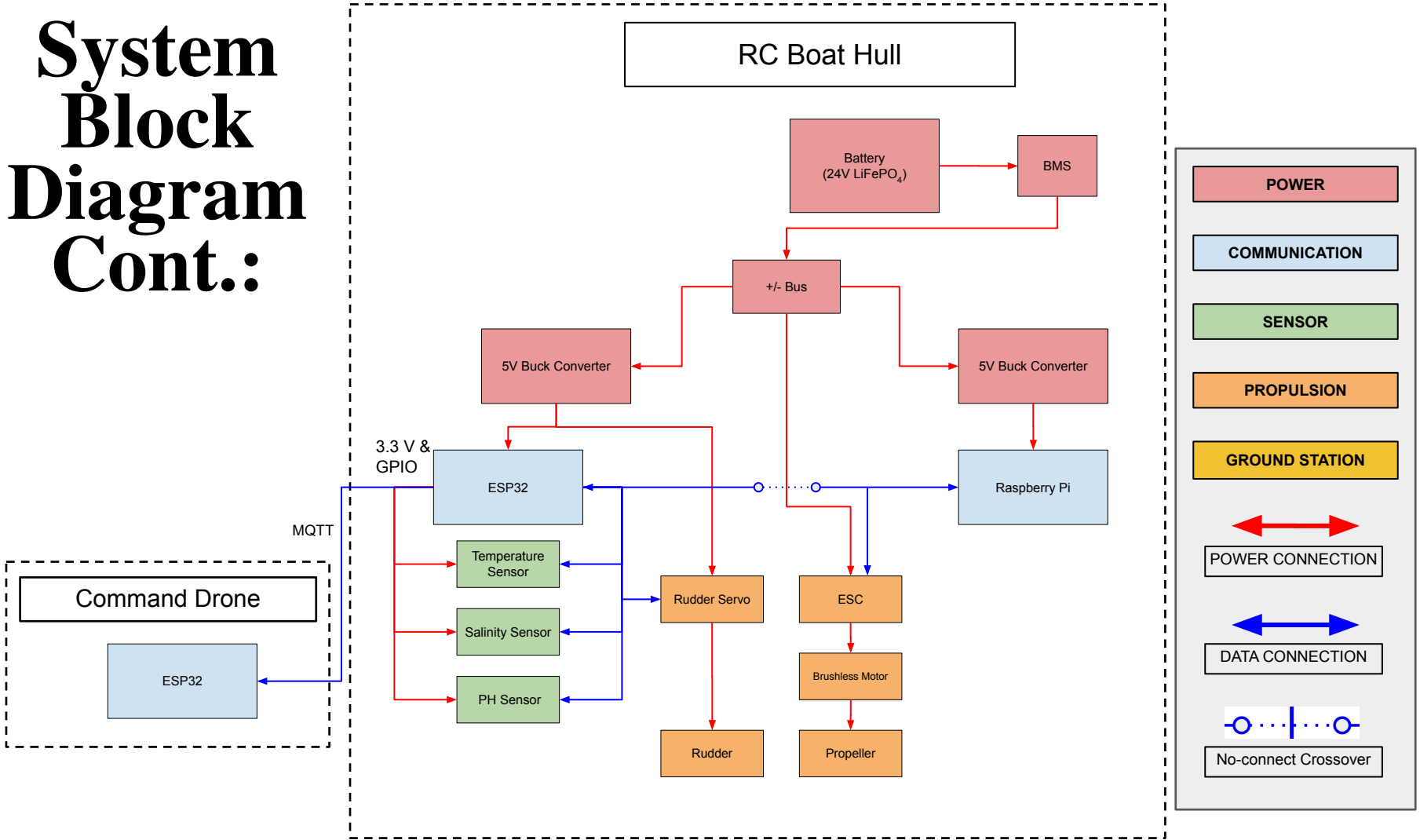
- Deployment system is no longer within the scope of the project
- Shift of project focus to showcasing swarm behavior and control
- At least three drones must be deployed
- Requirements including speed, battery life, and internal volume have been altered to include a threshold value and objective value

System Requirements

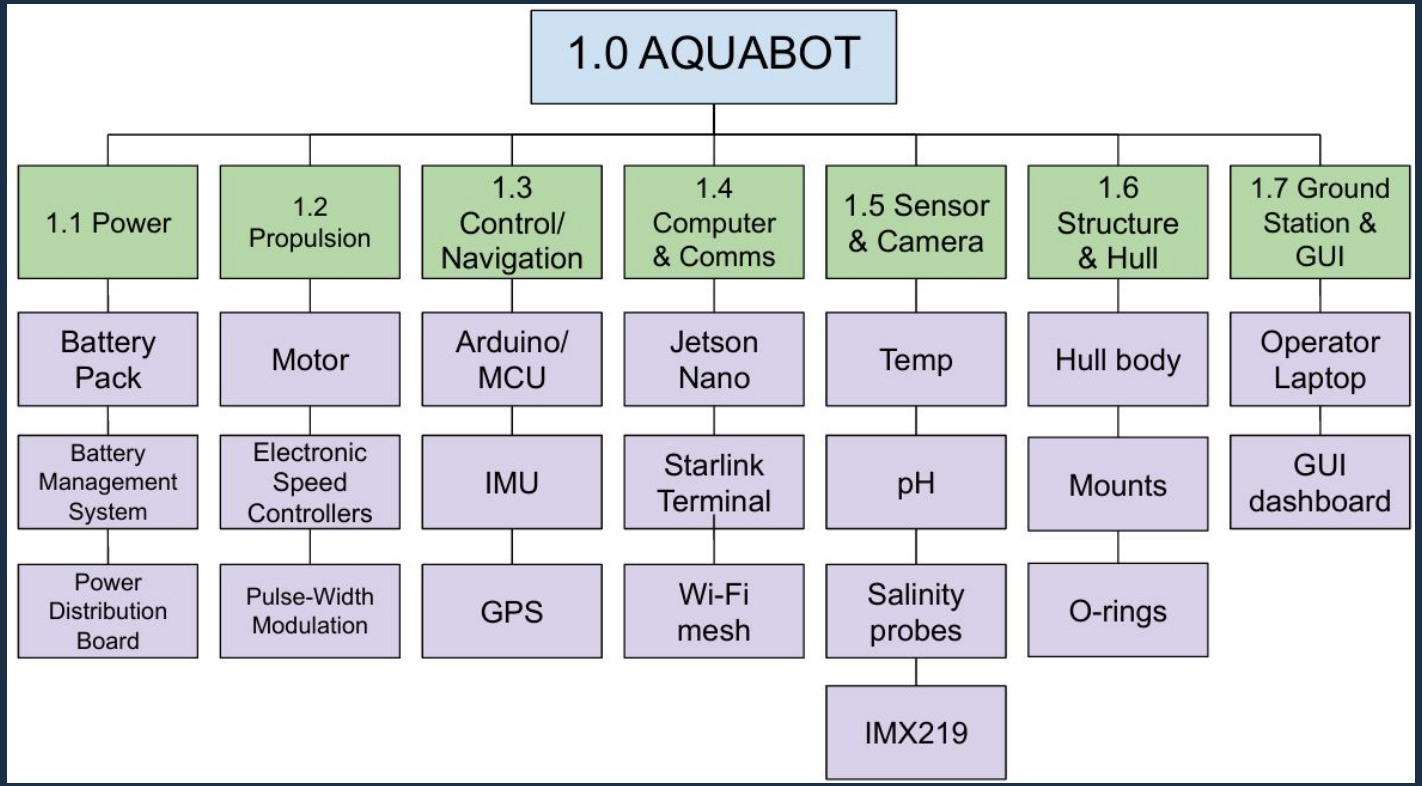
System Block Diagram:



System Block Diagram Cont.:



System Architecture



Key Performance Requirements

- **Swarm Deployment:** All drones shall autonomously self-right and enter operational state within 60s of impact.
- **Stability:** Each drone shall self-right and maintain operational orientation IAW UNE-EN ISO 12217-3:2022 and/or ASTM F3052-14(2020)e1.
- **Speed:** Each drone shall be able to **sustain a threshold speed of 2.5 kts** in calm water conditions, with an objective speed of 5 kts.
- **Battery Life:** Each drone shall reach a **threshold runtime of 6 minutes** at nominal load, with an **objective value of 5 hours.**
- **Data Collection:** Each drone shall measure temperature, salinity, pH and capture visual imagery at a minimum of 1080p resolution, 30 fps.
- **Data Mapping:** Each drone shall generate a georeferenced map of collected environmental variables and export data in GIS-compatible format.

System Requirements Verification Matrix (SRVM)

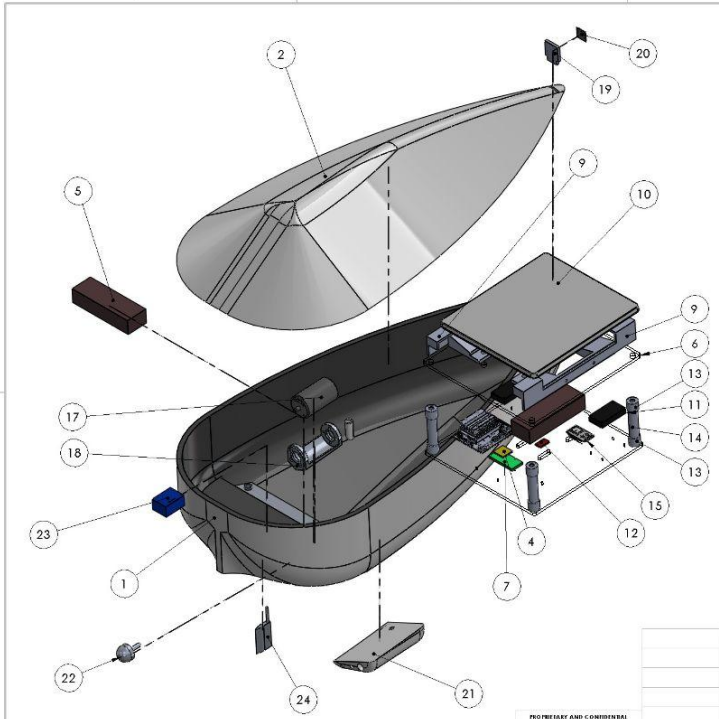
Requirement Type	Requirement	Verification Method			
		Test	Analysis	Demonstration	Inspection
4.1 Performance Requirements	4.1.2 Swarm Control			X	
	4.1.3 Stability	X		X	
	4.1.4 Speed	X			
	4.1.5 Battery Life	X	X		
	4.1.6 Data Collection	X		X	
	4.1.7 Data Mapping		X		
4.2 Design Requirements	4.2.1 Hull Shape		X		
	4.2.2 Internal Volume		PASS		PASS
	4.2.3 Materials				X
	4.2.4 Payload Mounts			X	X

SRVM Cont.

Requirement Type	Requirement	Verification Method			
		Test	Analysis	Demonstration	Inspection
4.3 Environmental Requirement	4.3.1 Survivability	X	X		
	4.3.2 Waterproofing	X			X
	4.3.3 Operating temperature	X	X		
	4.3.4 Non-Hazardous Surface				PASS
4.4 Interface Requirements	4.4.1 Autonomy & Control			PARTIAL PASS	
	4.4.2 Inter-Drone Communication	PARTIAL PASS		X	
	4.4.4 Satellite Uplink	PASS		PASS	
	4.4.5 Display			PARTIAL PASS	PASS

Point Design

Assembly Drawing



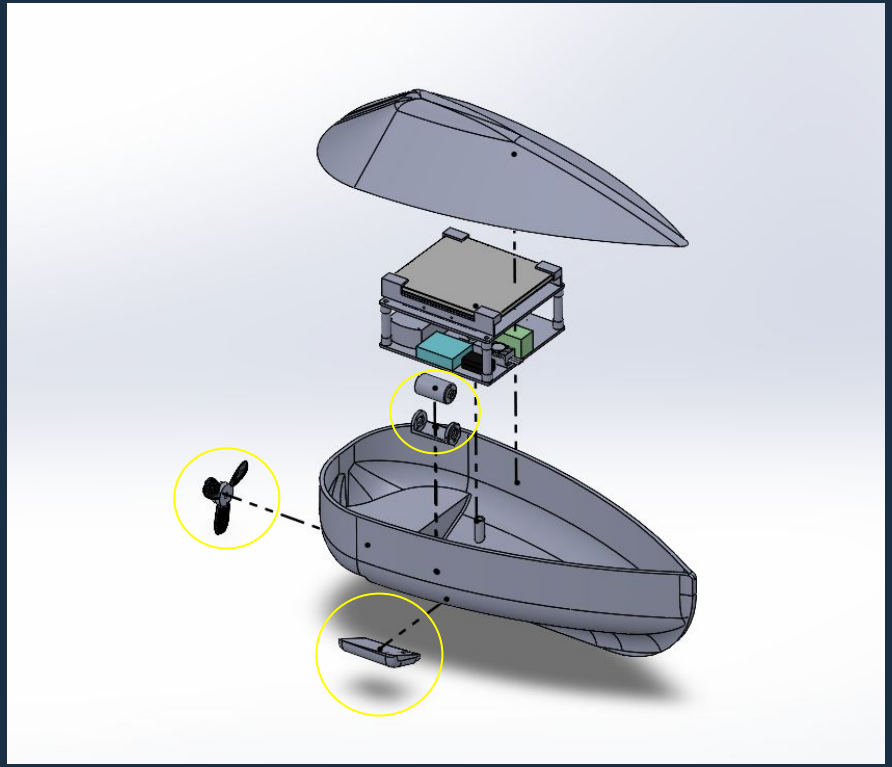
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	14012	Lower Shell	1
2	14011	Upper Shell	1
3	11000, 14020	Shelf System	1
4	AQ409_GPS-1	Gloal Positioning System	1
5	11060	Battery 1	2
6	TopShelf2.2	Shelf, Top	1
7	AQ415_RaspPI-1	Raspberry PI Cmputer	1
8	AQ405_StarlinkMountBase-1	StarLink Mount, Bottom	1
9	AQ406_StarlinkMountBracket-1		2
10	AQ407_Starlink-1	StarLink Mini	1
11	AQ412_StepDownConverter-1	Step Down Converter	2
12	AQ410_IMU-1	Inertial Measurement Unit	1
13	AQ413_ShelfSupportEndCaps-1	Support End Caps	8
14	AQ408_SupportColumns-1	Support Columns	4
15	AQ429_2023Esp32-1		1
16	AQ417_JetsonNano-1		1
17	AQ418_MainMotor-1		1
18	AQ420_MotorMount-1		1
19	AQ419_CameraHousing-1		1
20	AQ423_Camera-1		1
21	AQ425_ThinSensorMount2023-1		1
22	AQ426_PropellerScaled-1		1
23	AQ427_Servo-1		1
24	AQ428_Rudder-1		1

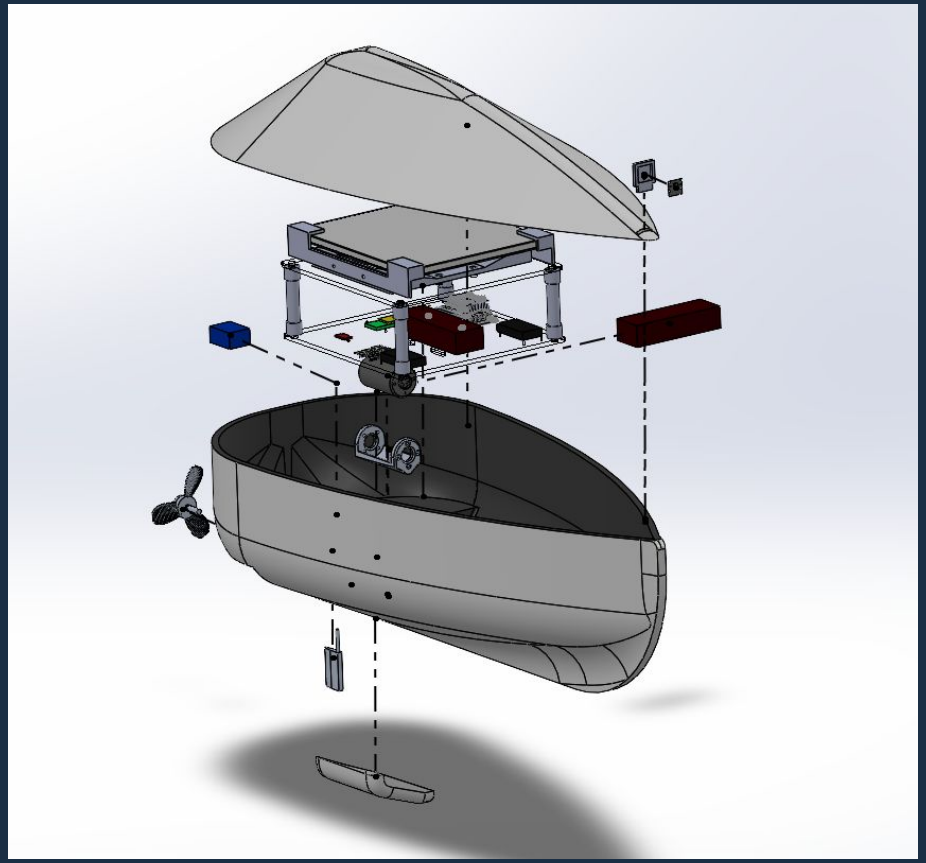
UNLESS OTHERWISE SPECIFIED: NAME DATE
DRAWN
CHECKED
ENG APPR.
MFG APPR.
Q.A.
COMMENTS:
TITLE:
SIZE DWG. NO. REV
AQ201_Aquabot-2
SCALE: 1:20 WEIGHT: SHEET 1 OF 1

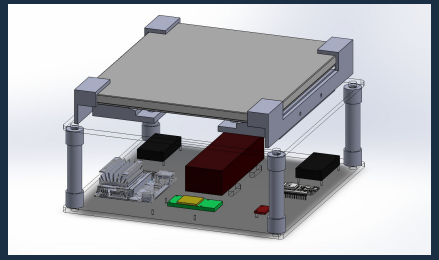
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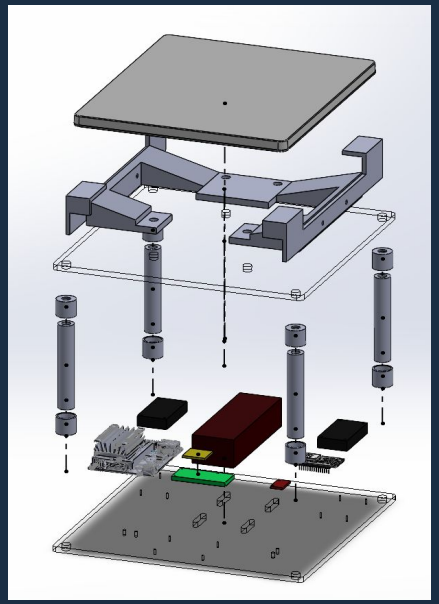
NEXT ASSY USED ON
APPLICATION DO NOT SCALE DRAWING

Exploded SolidWorks Assembly





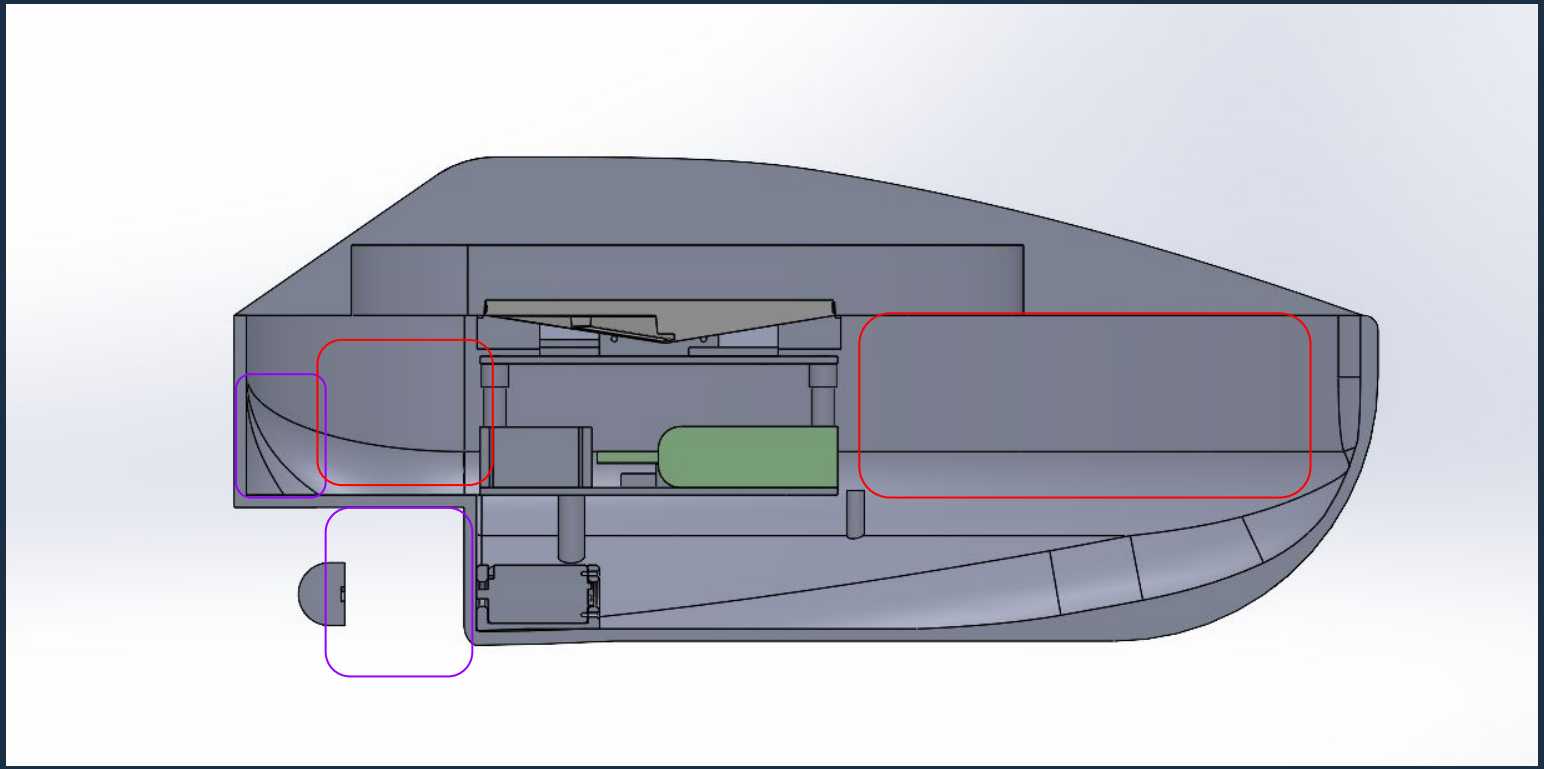




*Updated electronics:
- Starlink
- Jetson Nano
- ESP32
- Battery

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Exploded SolidWorks Assembly

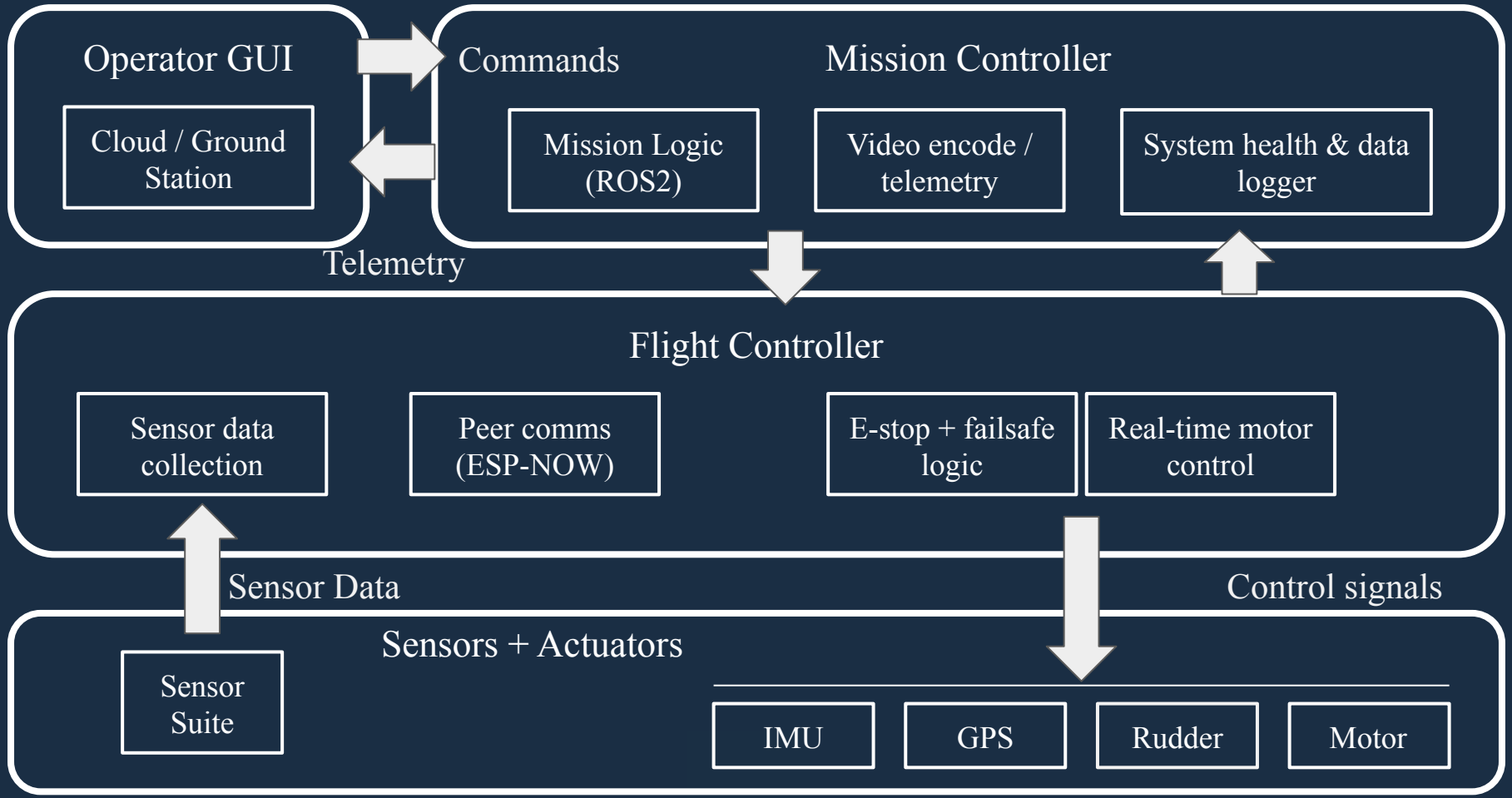


Parts Guide for CAD

Part Number	Part Name	Rev Number	Status
1	CompletePart	1	Complete
2	WIPPart	1	WIP
3	ObsoletePart	1	OBS
4	To-DoPart	1	To-Do
201	Aquabot	1	Complete
202	ShelfAssem	1	Complete
203	StarlinkMount	1	Complete
401	BottomShell	1	Complete
402	TopShell	1	Complete
403	BottomShelf	1	Complete
404	TopShelf	1	Complete
405	StarlinkMountBase	1	Complete
406	StarlinkMountBracket	1	Complete
407	Starlink	1	Complete
408	SupportColumns	1	Complete

Part Number	Part Name	Rev Number	Status
409	GPS	1	Complete
410	IMU	1	Complete
411	VerticalSupports	1	Complete
412	StepDownConverter	1	Complete
413	ShelfSupportEndCaps	1	Complete
414	ShelfSpacer	1	Complete
415	RaspPi	1	Complete
416	LiPO	1	Complete
417	JetsonNano	1	Complete
418	MainMotor	1	Complete
419	CameraHousing	2	Complete
420	MotorMount	1	Complete
421	Battery	1	Complete
422	RCMiniPiMount	1	Complete
501	GoPro	1	OBS

Software Subsystem


Operator GUI
Cloud / Ground Station
Commands
Telemetry
Mission Controller
Mission Logic (ROS2)
Video encode / telemetry
System health & data logger
Flight Controller
Sensor data collection
Peer comms (ESP-NOW)
E-stop + failsafe logic
Real-time motor control
Sensor Data
Control signals
Sensors + Actuators
Sensor Suite
IMU
GPS
Rudder
Motor

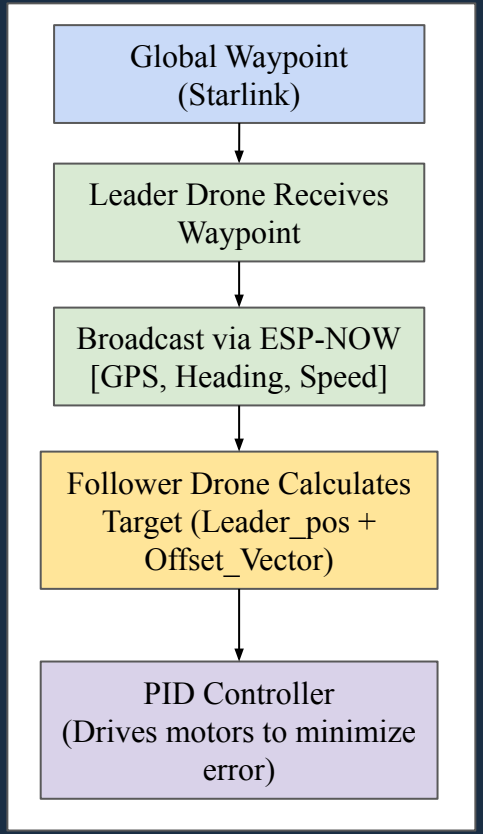
Swarm Architecture

Decentralized Mesh Network (Communication)

- **Protocol: ESP-NOW** (IEEE 802.11 Action Frames). Operates at Layer 2 (MAC) for connectionless, router-free communication.
- **"Any-to-Any" Routing:** Implements a custom **Flooding Algorithm** with TTL=3 (Time-To-Live).
  - *Logic:* Packets are re-broadcast by neighbor drones, creating dynamic relay paths if the direct Line-of-Sight is blocked by waves.
- **Performance Verification:**
  - **Latency: < 10 ms** (Verified). This is 10x faster than the system reaction time requirement, enabling real-time collision avoidance.
  - **Throughput:** 50 packets/sec verified (Sufficient for 10 Hz control loop).

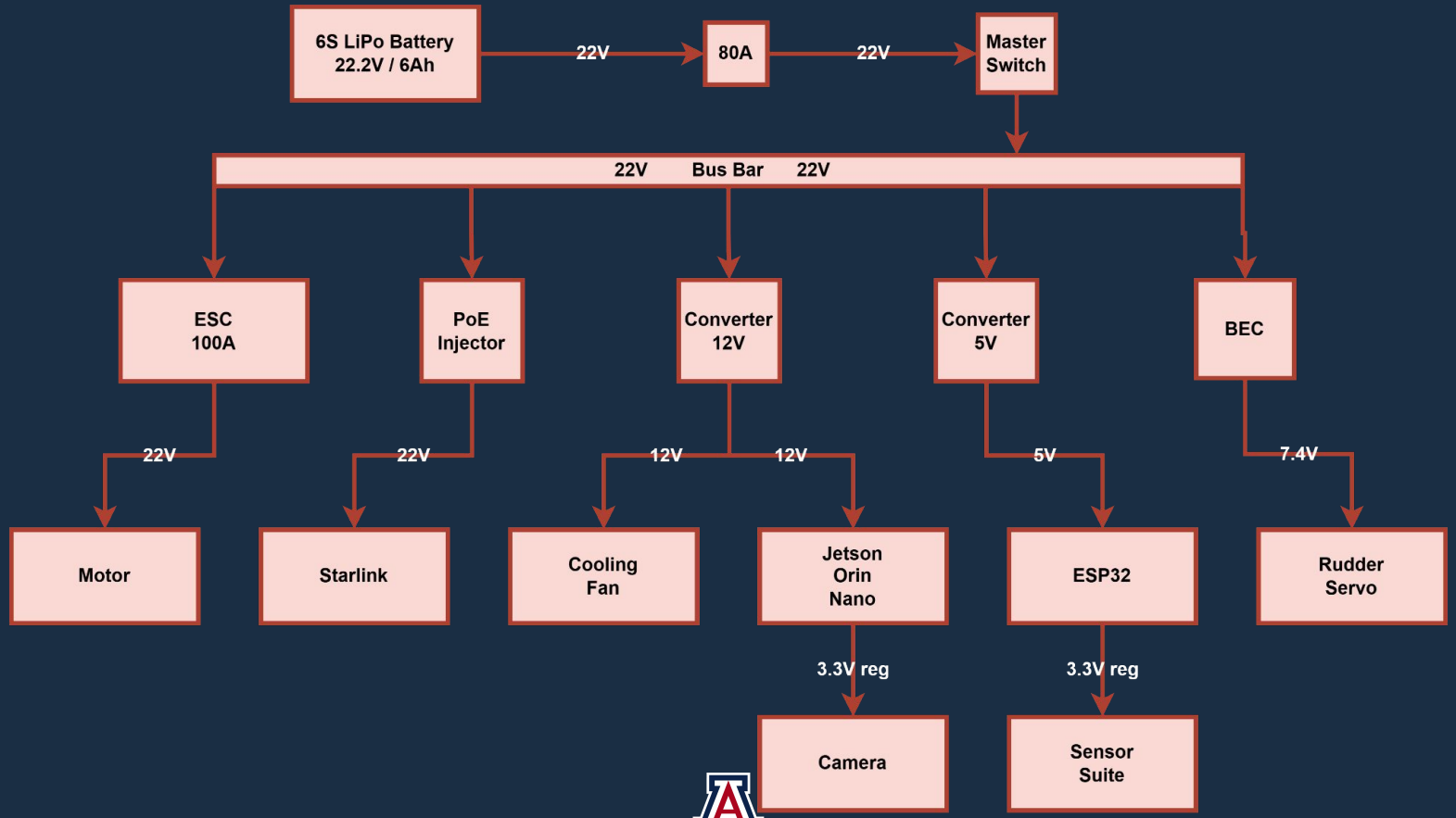
Content Block 2: Formation & Control Logic (Autonomy)

- **Strategy: "Virtual Rigid Body"**. The swarm treats the Leader as the origin point (0,0) and moves as a single fixed coordinate frame.
- **Position Algorithm:**
  - Follower derives target using: $Target_{GPS} = Leader_{GPS} + R(\theta) \times Offset$
  - *(Where* $R(\theta)$ *is the Leader's heading rotation matrix).*
- **Control Loop:**
  - **PID Controller:** Drives motor thrust to zero out the Distance_Error.
  - **Failsafe:** If data age > 3.0s, logic switches to **"Station Keep"** (Hold GPS Position) until the mesh heels.

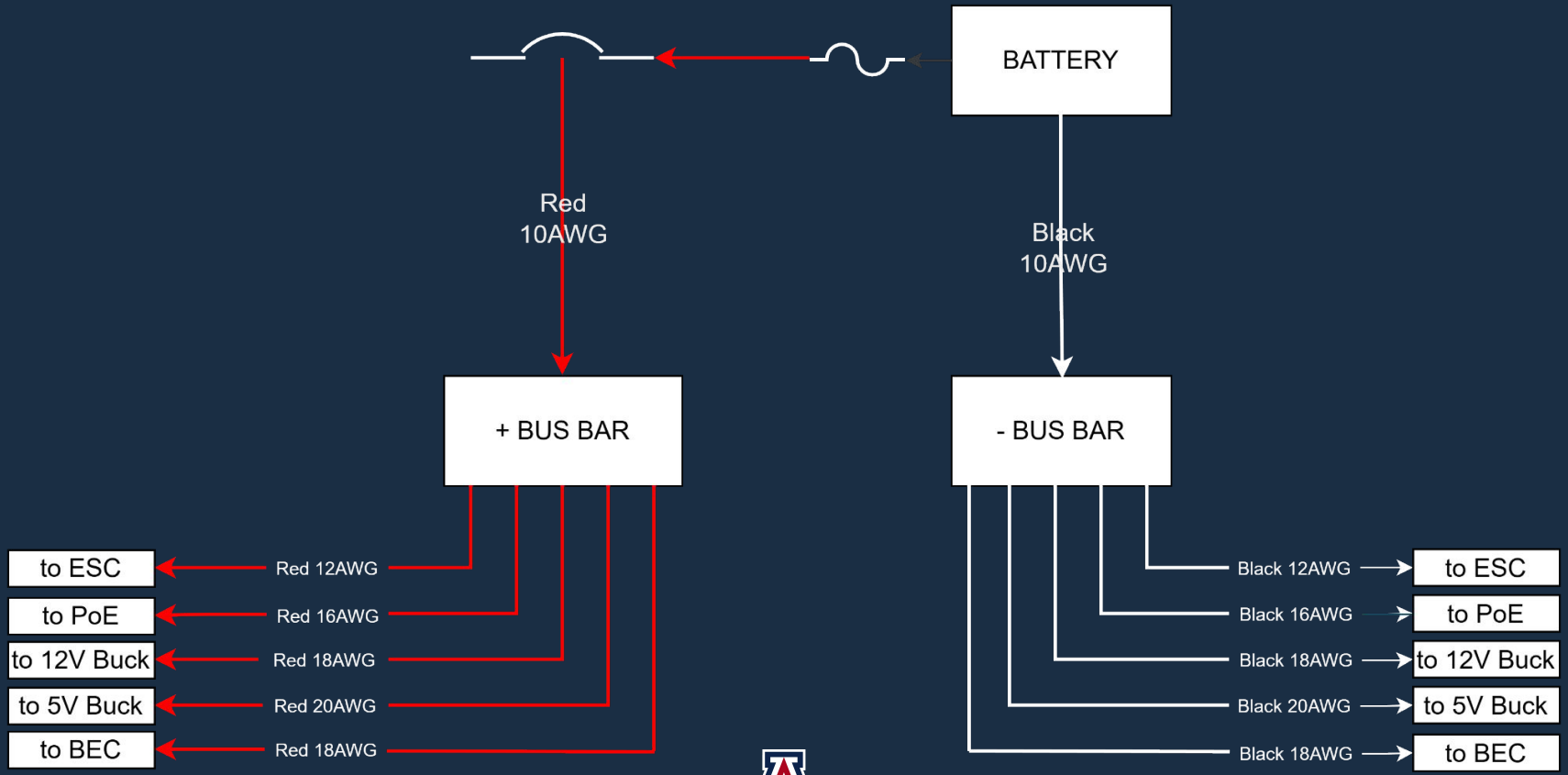


Power Subsystem

Block Diagram



Wiring Diagram



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Power Budget

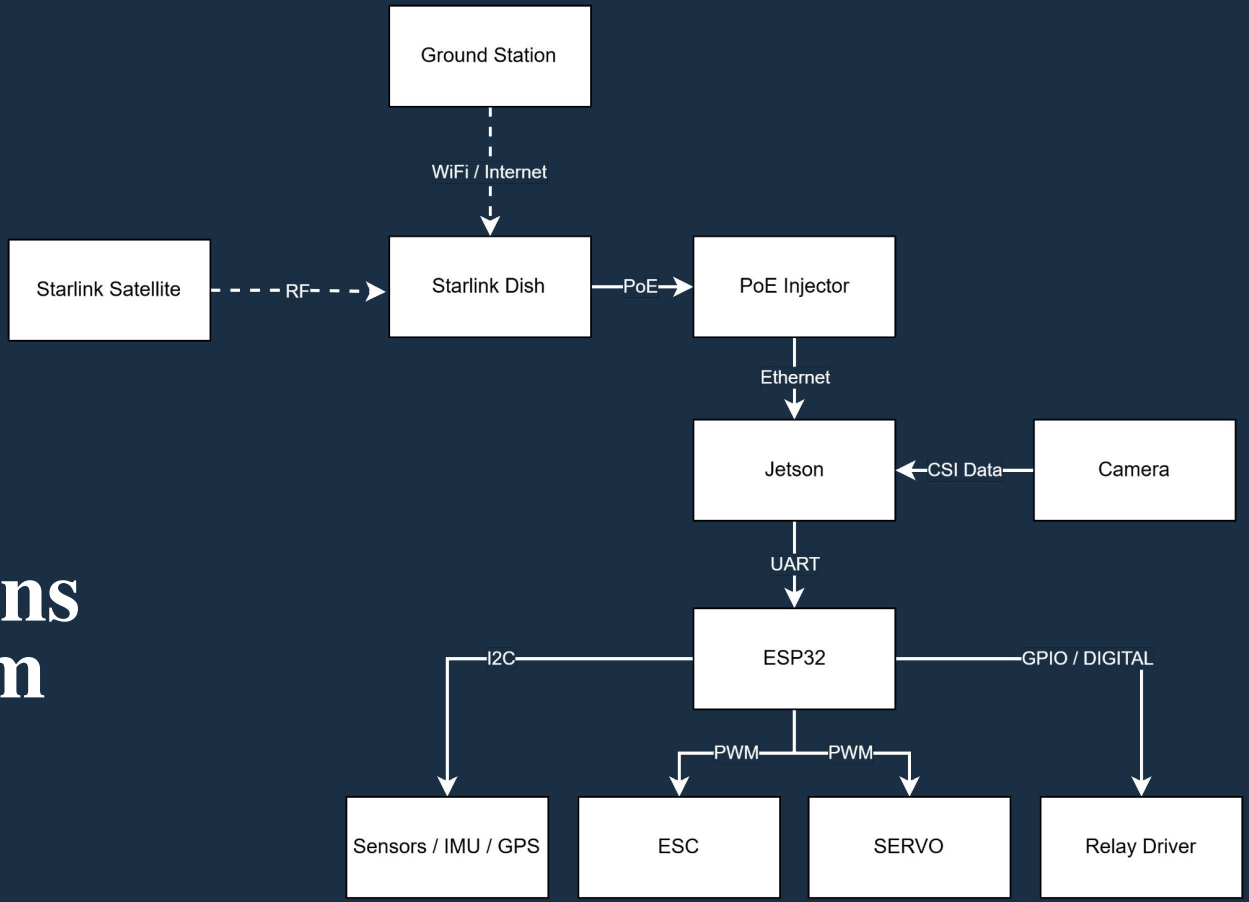
Subsystem	Component	Voltage Rail	Max Power (Load)	Efficiency Loss	Total Power (from Battery)	Max Current (from Battery)	Cruise Power (from Battery)
Subsystem	Component	Voltage Rail	Max Power (Load)	Efficiency Loss	Total Power (from Battery)	Max Current (from Battery)	Cruise Power (from Battery)
Propulsion	Apisqueen 5080 Motor	22.2 V	1200 W	0 W (Direct)	1200.0 W	54.0 A	60.0 W
Comms	Starlink Gen 3 (via Yaosheng)	57 V	100 W	~5 W (95%)	105.0 W	4.7 A	70.0 W
Compute	Jetson Orin Nano + Fan	12 V	25 W	~3 W (90%)	28.0 W	1.3 A	15.0 W
Control	ESP32 + Rudder Servo	5V / 7.4V	20 W	~2 W (90%)	22.0 W	1.0 A	5.0 W
Payload	Sensors (GPS, IMU, Env)	3.3 V	5 W	~1 W (85%)	6.0 W	0.3 A	5.0 W
TOTALS			**1350 W**		**~1361 W**	**~61.3 A**	**~155 W**

Battery Capacity:	**Ovonic 6S LiPo (22.2V, 6000mAh) = 133.2 Wh**
Cruise Endurance:	**133.2 Wh / 155 W ≅ 0.85 Hours (51 Minutes)**
Safety Margin:	**Max draw is 61.3A, well within the 80A main fuse and the battery's 120C (720A) discharge limit.**

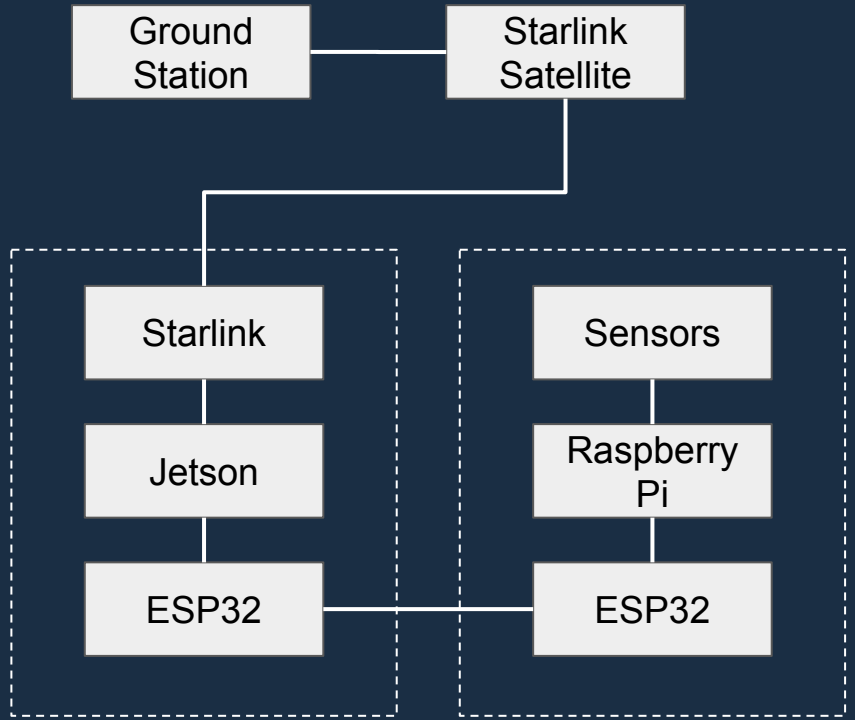
Communication Subsystem



Communications Block Diagram



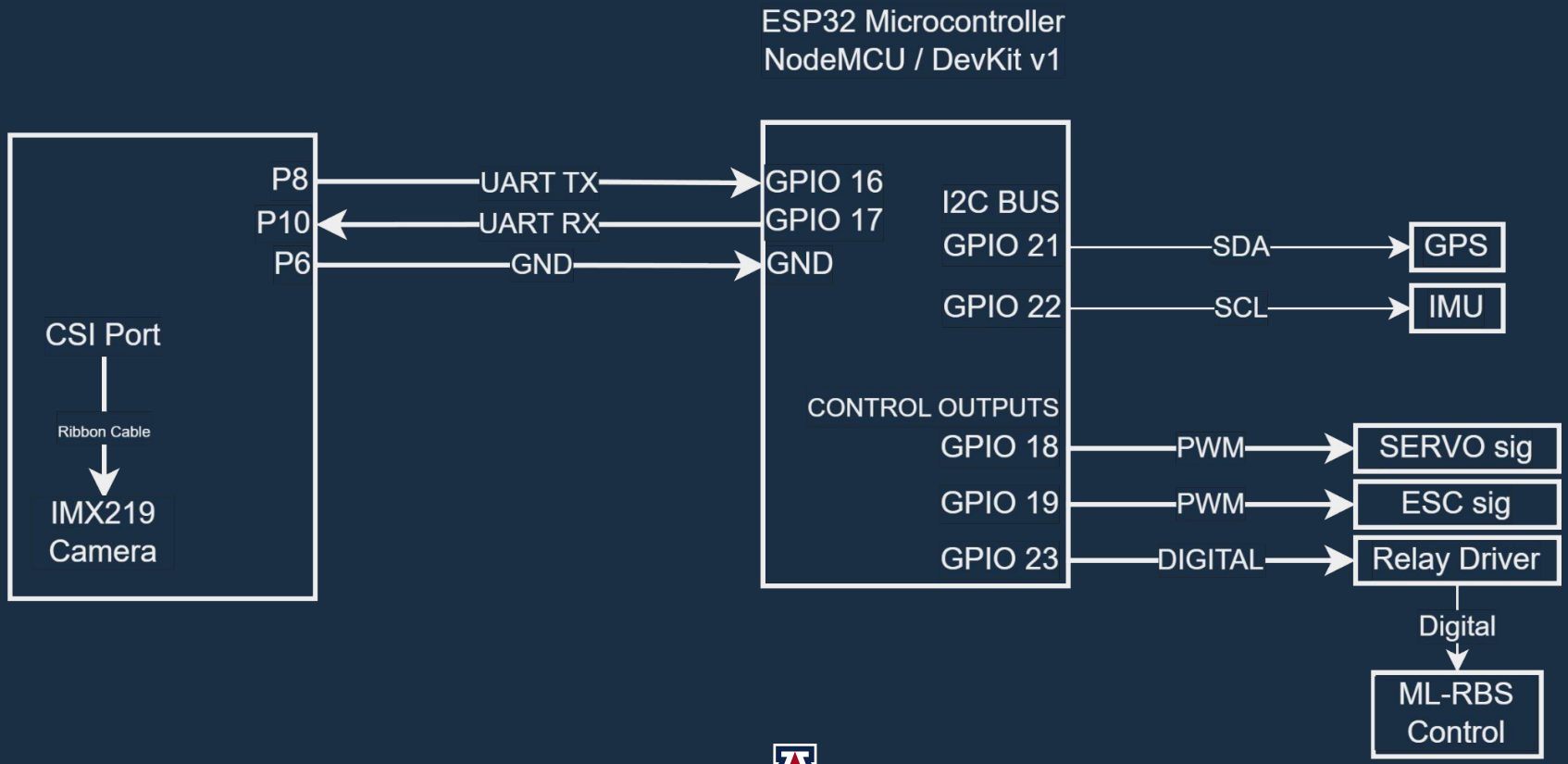
Gateway and Swarm Topology



Hybrid Star-Mesh will utilize a "Gateway & Swarm" architecture to minimize cost while maximizing range.

Feature	Main Drone (Gateway)	Follower Drones (Swarm)
Role	Central Hub & Internet Bridge	Data Collection Node
Uplink	Starlink (Marine Terminal)	None (Relays to Main)
Local Link	ESP-NOW (Rx/Tx)	ESP-NOW (Tx to Main)
Hardware	Jetson Nano + ESP32	Raspberry Pi + ESP32
Data Flow	Aggregates all swarm data → Cloud	Sensors →Main Drone

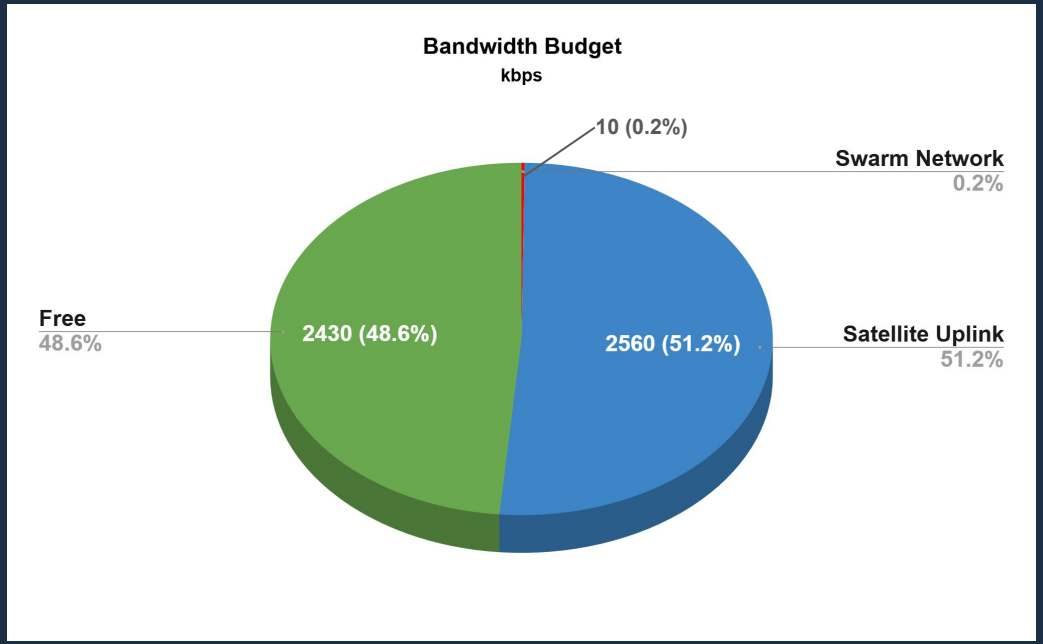
Wiring Diagram



Performance & Bandwidth Analysis

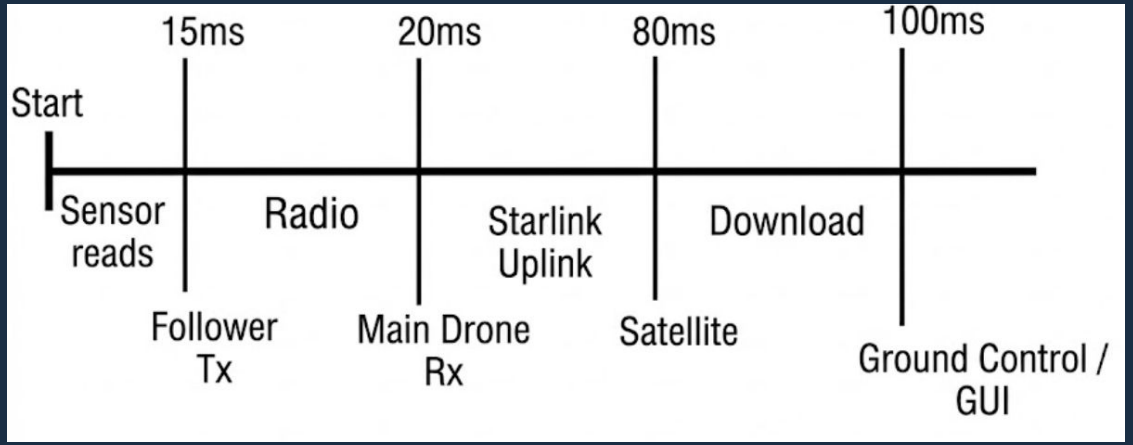
Link Budget

Link Type	Available Capacity	Required Load	Margin / Status
Swarm Network (ESP-NOW)	~1,000 kbps	10 kbps (2 Drones)	**99% Free (Highly Scalable)**
Satellite Uplink (Starlink)	~5,000 kbps	2,560 kbps (Video + Telemetry)	**~50% Free (Video Quality Safe)**



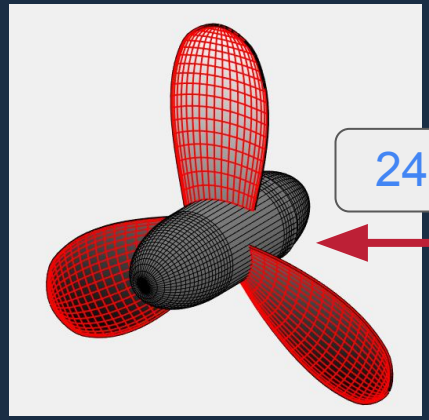
Data Protocol & Latency

- **Protocol Stack Definition** We utilize distinct protocols for the local swarm versus the global uplink to optimize for speed and reliability.
    - **Swarm Link: ESP-NOW**
    - **Video Uplink:** WebRTC
    - **Telemetry:** MQTT
- **Data Payload (The Packet)**
    - **Format:** Packed Binary Struct (64 Bytes).
    - **Content:** [Nav Data] + [Sensor Readings] + [Battery Status].
    - **Efficiency:** 10x smaller than standard JSON, reducing airtime collision risk.



Propulsion Subsystem

Propeller

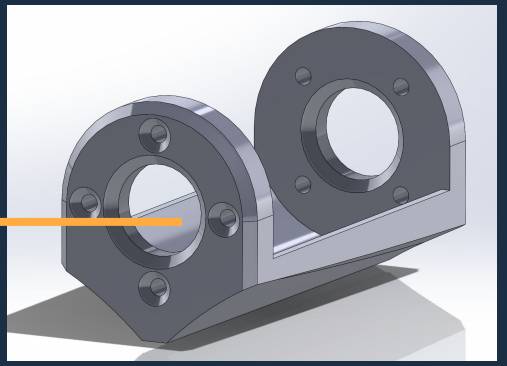


2400 RPM

Motor



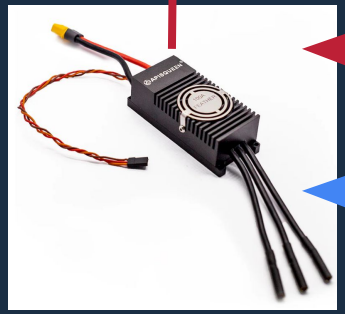
Motor Mount



24V

24V

Power from Battery



Control signal from ESP32

ESC

Data Connection

Power Connection

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RC Drone Specifications





Key Features:

- Can hold ESP32 and Jetson Nano for data transmission
- Reaches 45 km/h
- Waterproof
- Brushless Motor
- Watercooled System
- 190 min charging time
- 7 minute run time

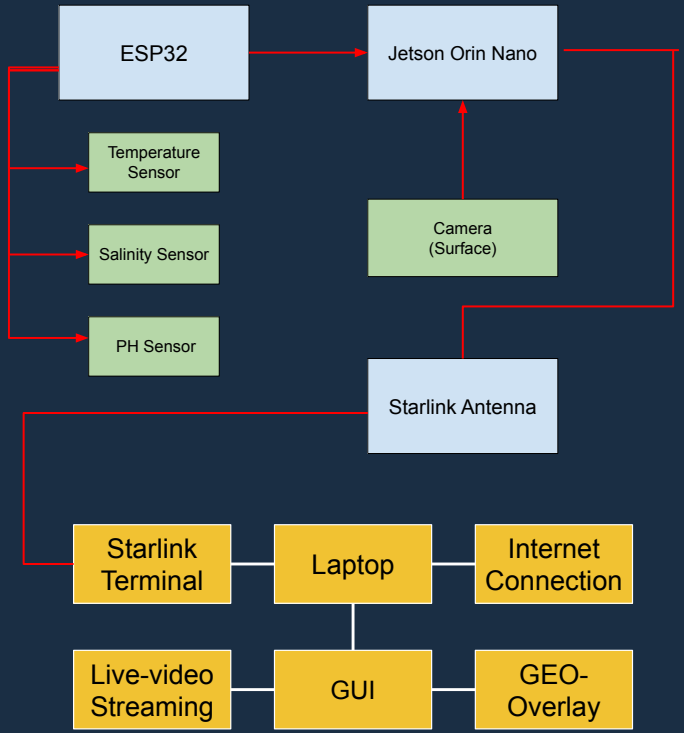
Notes:

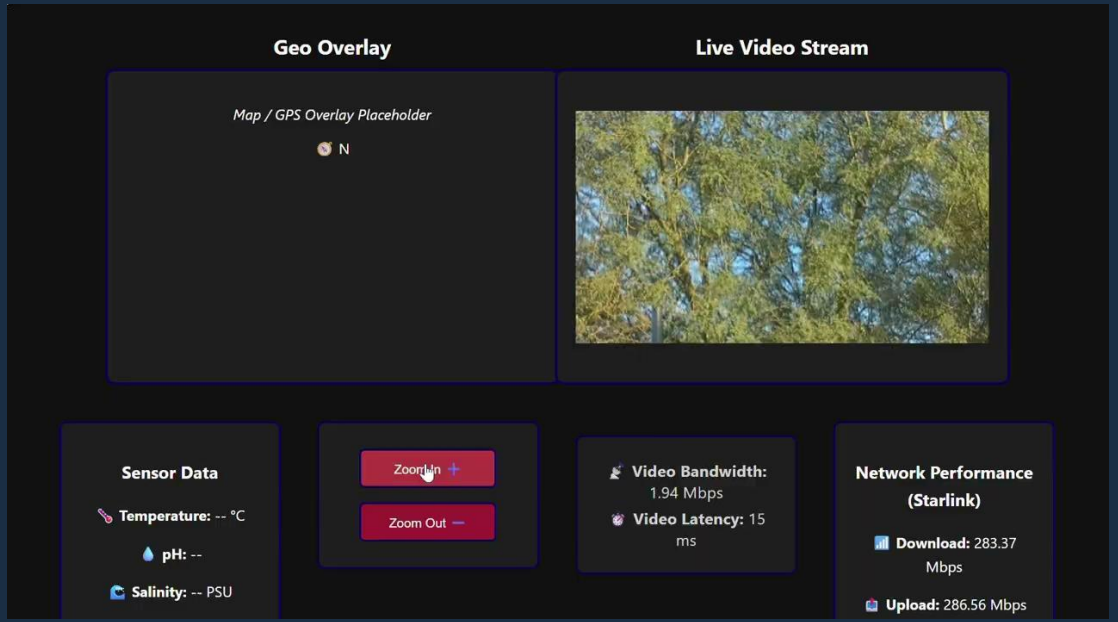
Used purely to demonstrate swarm abilities.
$1700 cheaper than using 2025 hull

Images from
https://www.amazon.com/-/es/Ssccgym-UDI005-escobillas-funciones-completas/dp/B0DRNNY597

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GUI with Sensor Data and Live Footage





Performance Analyses and Expectations

Propulsion Expectations

PDR:

Target cruise speed: 5 kn

CDR:

Target cruise speed: 2.5 kn

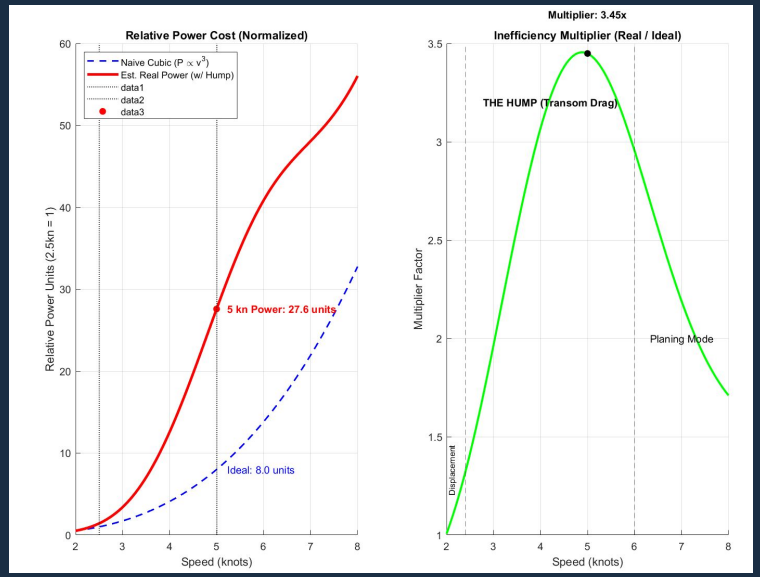
Hull Speed: *Maximum* speed a hull can reach before the bow wavelength equals the boat's length
→ Very large increase in wave drag

$$v_{hull} \approx 1.34\sqrt{L_{WL}}$$

L_WL is the waterline length in feet

v_hull (3.2 ft.) = 2.4 kn

Propulsion Expectations



$$P_{ideal}(v) = P_{ref} \cdot \left(\frac{v}{v_{ref}}\right)^3$$

$$P_{real}(v) = P_{ideal}(v) \times \mathcal{M}(Fn)$$

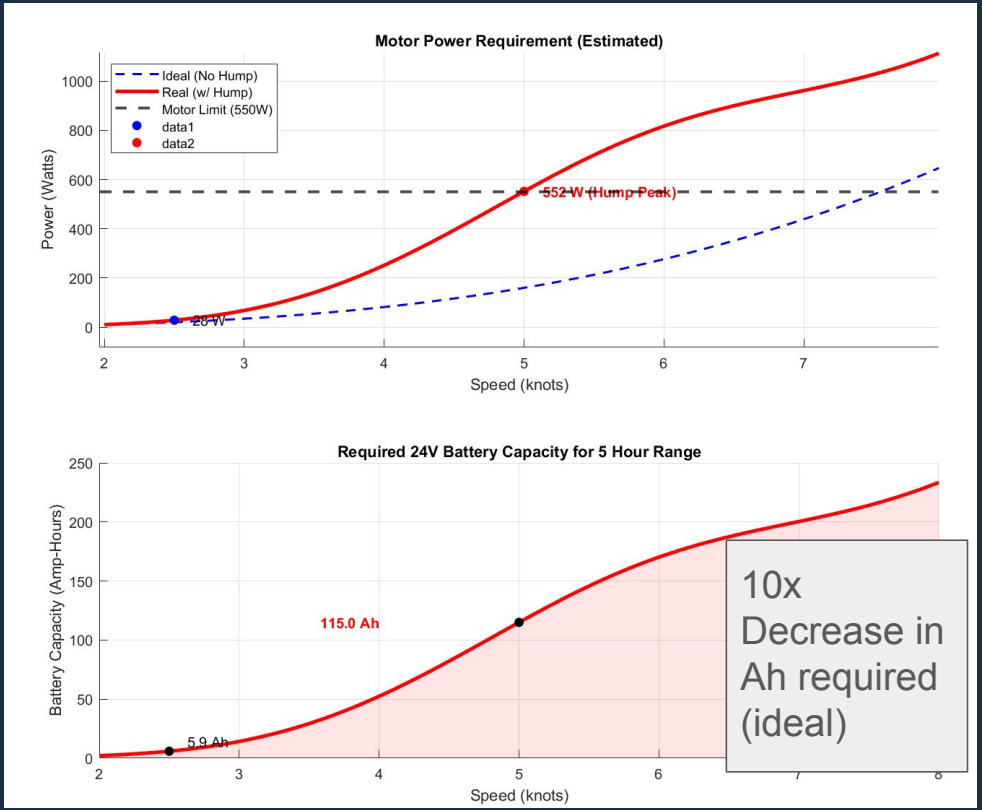
$$Fn\text{: Froude Number} = \frac{v}{\sqrt{g \cdot L}}$$

$$\mathcal{M}(Fn) = \underbrace{[1 + 0.5\tanh(5(Fn - 0.5))]}_{\text{Base Planing Shift}} + \underbrace{2.0 \cdot e^{-\frac{(Fn-0.8)^2}{2(0.25)^2}}}_{\text{Wave Hump Penalty}}$$

Accounting for wave drag (2.5 to 5 kn)

- 8x increase (ideal) vs ~28x increase (real)

Propulsion Expectations



Top graph: Power to motor to achieve cruise speed

- Difference between ideal and real model predictions

Bottom graph: Battery capacity required to achieve 5 hour range

- Only ‘motor range’
- Ideal case, assuming propeller efficiency = 1
- **115 Ah** battery for 5 kn
- **6 Ah** battery for 2.5 kn
  - Double battery capacity for same distance traveled: 12 Ah total

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Heat Analysis

Assumptions:

- AQUABOT modeled as two hemispheres and a cylinder (a large curved object for external convection)
- Small Jetson Orin fan forces internal convection slowly, comparable to flow over a flat plate
- AQUABOT operating stagnant, at steady state
- Radiation neglected
- Each wall layer is modeled as thin
- Standard values for conductive coefficients of dense fiberglass and marine rubber/plastic coating

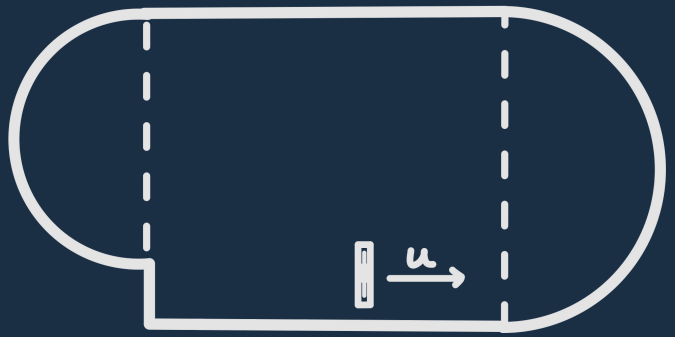


Image of AQUABOT Hull Approximated Geometry

$$\Rightarrow \left\{ \begin{array}{l} Re = \frac{uL_c}{\nu} \\ Nu \sim 0.664 Re^{\frac{1}{2}} Pr^{\frac{1}{3}} \\ h_i = \frac{Nuk}{L_c} \end{array} \right. \Rightarrow h_i \sim 5.45\ W/K * m^2$$

Heat Analysis

Heat from Electronics	Internal Heat Generation (W)	100
Approximate Geometry (as 2 hemispheres and a middle cylinder)	Rear Hemisphere Radius (m)	0.345
	Rear Gap Height (m)	0.15
	Cylinder Radius (m)	0.42
	Cylinder Length (m)	0.21
	Front Hemisphere Radius (m)	0.42
Material Properties	Total Thickness (m)	0.012
	Fiberglass Thickness (m)	0.01
	Coating Thickness (m)	0.002
	Fiberglass Conductivity (W/m*K)	0.3
	Coating Conductivity (W/m*K)	0.2
Convective Properties	Inside Air Convectivity (W/m^2*K)	5.45
	Outer Air Convectivity (W/m^2*K)	7
	Outer Air Temperature (K)	299
	Water Convectivity (W/m^2*K)	100
	Water Temperature (K)	297
Area Calculations	Rear Hemisphere Inner Surface Area (m^2)	0.6967
	Cylinder Inner Surface Area (m^2)	0.3999
	Front Hemisphere Inner Surface Area (m^2)	1.0459
	Total Inner Surface Area (m^2)	2.1426
	Rear Hemisphere Outer Surface Area (m^2)	0.7479
	Cylinder Outer Surface Area (m^2)	0.4130
	Front Hemisphere Outer Surface Area (m^2)	1.1084
	Total Outer Surface Area (m^2)	2.2692
Thermal Resistance Calculations	Inner Convective Thermal Resistance (K/W)	0.0857
	Conductive Thermal Resistance (K/W)	0.0200
	Outer Convective Thermal Resistance (K/W)	0.0337
	Total Thermal Resistance (K/W)	0.1394

Table with Heat Transfer Parameters and Calculated Values

$$R_{tot} = \frac{1}{h_i A_i} + \frac{\delta_1}{k_1 A_i} + \frac{\delta_2}{k_2 A_o} + \frac{1}{h_o A_o}$$

$$T_i = QR_{tot} + T_\infty$$

Conservative Estimate:

Internal Temperature (C)	42
Internal Temperature (F)	108

- Internal convection coefficient dominates behavior
- Further testing with cooling fan is required for more accurate estimation of internal flow speed
- Overall, final temperature is towards high end of electronics operating threshold

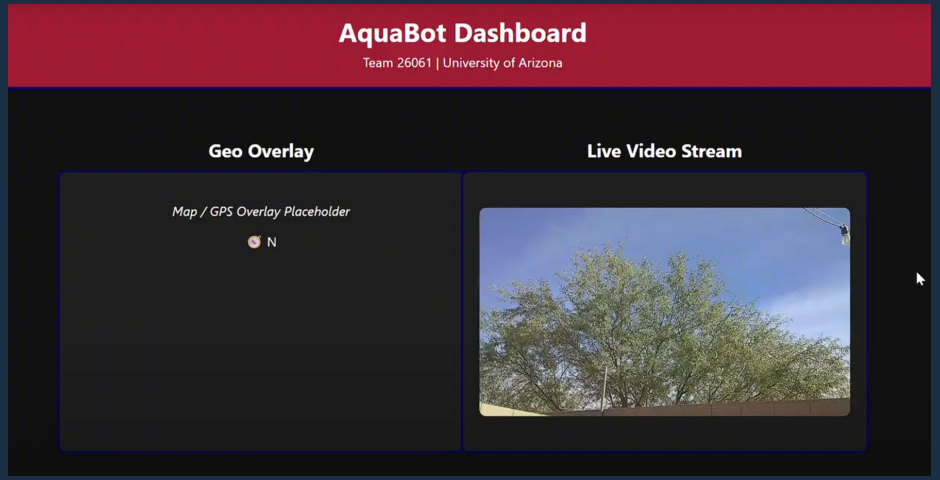
GUI and Ground Station Data Expectations

Sensor Data:

- All data syncs every 10 seconds (minimizes bandwidth used for data transmission)
- Temperature and GPS data integrated to create GEO-overlay for data visualization
- CSV file viewable to show location-based data for temperature, pH, and salinity.
- Saved onboard and on ground station GUI

Camera Data:

- Saved onboard and on ground station
- Livestream on GUI dashboard, with saved footage accessible



Verification

Verification Overview

Step 1: Verify Individual Function of All Electronics:

- Starlink✅
- ESP32✅
- Jetson Nano✅
- Sensors ☑️
- IMX219 (Camera)✅
- Motors
- Servos

Step 2: Verify Individual Function of All Electronics Inside Hull & Hull Qualities:

5.1.3 Stability: Each drone shall self-right and maintain operational orientation.
5.3.2 Waterproofing: Each drone shall meet IP68 ingress protection standards. Components shall be tested in 1.5 m depths to ensure compliance.

Step 3: Verify Swarm Capabilities

5.1.2 Swarm Control: All drones shall self-right and enter operational state in 60s.
5.4.1 Autonomy & Control: Each drone shall support autonomous navigation and remote operator control.
5.1.4 Speed: Each drone shall sustain a threshold speed of 2.5 kts in calm water conditions, with an objective speed of 5 kts.

✅= Fully Verified, ☑️ = Partially Verified

Detailed Verification Plan

5.4.2 Verification Procedure: Mesh Network Relay Test

- **Objective:** Verify that the ESP-NOW Managed Flooding Mesh successfully routes telemetry from a "Follower" drone to the "Leader" drone via an intermediate "Relay" drone when a direct link is unavailable.
- **Requirement Verified:** 5.4.2 Inter-Drone Communication (Multi-hop capability).
- **Success Criteria:** Leader receives telemetry packets from Follower with a latency of < 200 ms despite no direct Line-of-Sight (LOS).

Step	Action	Expected Result	Pass/Fail
1	Power on all 3 nodes (Leader, Relay, Follower) in close proximity (1m).	Leader Serial Monitor shows telemetry from "Follower" (Direct Link). RSSI > -50 dBm.	☐
2	Move "Follower" node to a distance of 100m OR place inside a metal RF shield, leaving only the "Relay" node in range.	Direct link is broken. Leader temporarily stops receiving Follower data.	☐
3	Position "Relay" node at the midpoint (150m) between Leader and shielded Follower.	The Mesh Logic activates. Leader resumes receiving telemetry from Follower.	☐
4	Inspect the received packet header on the Leader.	Packet shows Hop_Count >= 1 (indicating it passed through a relay).	☐
5	Measure Latency (Time Sent vs. Time Received).	Latency remains < 200 ms for the hopped packet.	☐

Detailed Verification Plan

5.1.6 Verification Procedure: Data Collection Test

- **Objective:** Verify that all temperature, salinity, and pH data collected is reliably saved onboard and transmitted to ground station, as well as camera footage being at 1080p, 30 fps.
- **Requirement Verified:** 5.1.6 Data Collection.
- **Success Criteria:** All data collected is accurate and adequately displayed

Step	Action	Expected Result	Pass/Fail
1	Test all three sensors in a liquid of known qualities. (Not transmitted via Starlink)	Data collected matches known values	☐
2	Retake data with transmission to GUI via Starlink	GUI data matches known values	☐
3	Connect camera to GUI and onboard storage, inspect saved data.	Saved data matches and is accessible via GUI	☐
4	View stored data on ESP32 via connection to monitor and stored data on GUI	Stored data demonstrates equivalent values per timestamp	☐
5	Test and modify GUI for usability and interpretation of data	GEO-overlay accurately displays data in a user-friendly manner, all data is easily accessible. Camera live stream and review is functional.	☐

Detailed Verification Plan

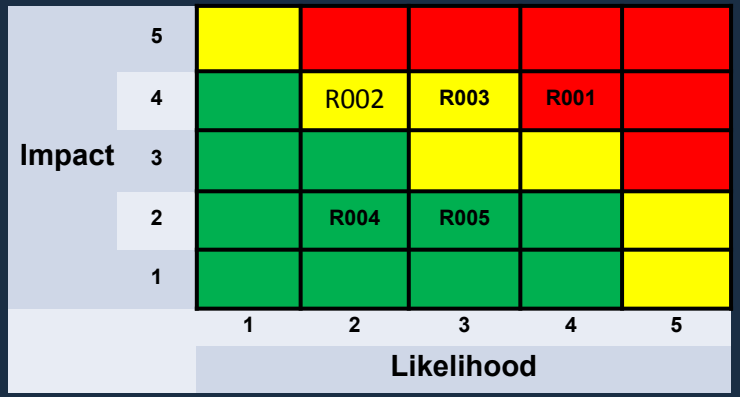
5.4.6 Verification Procedure: Swarm Control

- **Objective:** Successfully communicate position data and maintain formation within the specified timing constraints (<1s latency & >= 1Hz update)
- **Requirement Verified:** 5.4.6 Swarm Control
- **Success Criteria:** The swarm shall maintain coordinated network behavior through a leader-follower model executed via Jetson Mission Computer and MQTT topics. Each drone shall publish and subscribe to swarm state messages at ≥ 1 Hz with < 1 s latency.

Step	Action	Expected Result	Pass/Fail
1	Initialize Swarm Network: Each drone connects to the MQTT broker, and verifies communication with the Jetson Nano.	Ground station receives communication from both Drone IDs via MQTT; Link Quality > 50%	☐
2	Establish Leader-Follower Roles: The leader drone publishes its state and navigation cues, while follower drones sync.	Follower drones successfully identify the Leader ID, switch to *following* state, and begin mirroring the Leader's heading and velocity in 1s or less.	☐
3	Exchange Swarm State Messages: All drones publish and subscribe to swarm state topics at ≥1 Hz, ensuring message latency stays under 1 s.	Telemetry logs confirm that packets form every Drone ID appear in the swarm/state topic at intervals of <= 1s (>= 1Hz). Timestamp shows the delay between *sent* and *received* time is consistently less than 1s.	☐
4	Coordinate Navigation: Using GPS data, each drone adjusts its path to maintain formation and avoid collisions while following trajectory.	Follower drones autonomously adjust motor thrust/yaw to maintain target separation distance from the Leader. Visual observation and GPS tracks confirm drones move in parallel paths **without** colliding.	☐
5	Collect Data: Each drone gathers environmental data and publishes it to the swarm network for processing.	Ground station compiles a unified dataset containing sensor readings from all distinct Drone IDs. The aggregated file shows no data gaps larger than 3 seconds and timestamps are synchronized	☐

Risk Analysis

Risk Analysis



	High Risk
	Moderate Risk
	Low Risk

Impact	Likelihood
5 - Catastrophic	5 - Near Certainty
4 - Major	4 - Highly Likely
3 - Moderate	3 - Likely
2 - Minor	2 - Somewhat Likely
1 - Insignificant	1 - Not Likely

ID	Risk Description	Severity (L x I)
R001	Integrating all software components	16
R002	Hull fragility due to modification	8
R003	Lack of software knowledge	12
R004	Assessing capabilities of smaller drones	4
R005	Modifying pre-manufactured drones	6

Risk Analysis: Mitigation Plan

R001 (high): Integrating all software components

1. Consult with experts and use university resources
    a. Find out where problems lie early
2. Plenty of tests to ensure successful results
    a. Correct input-output mapping

R002 (medium): Hull fragility due to modification

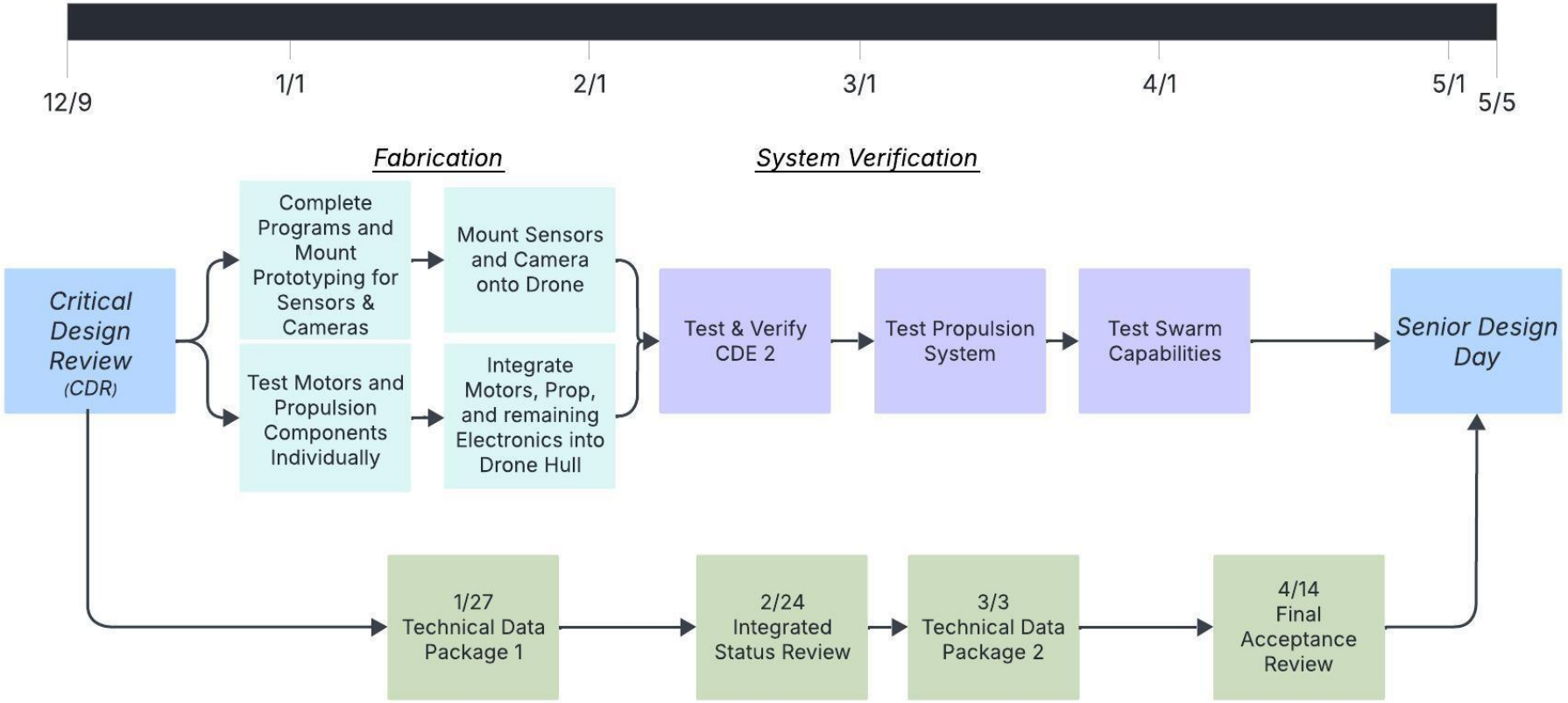
1. Talk to Eddie's Marine Equipment to learn correct modification methods
2. Investigate alternative mounting solutions (no hull modification)
3. Conduct isolated hull testing post-modifications

R003 (medium): Lack of software knowledge (we don't know what we don't know)

1. Use of AI as a guide to find surface level flaws
2. Consult with experts and use university resources to further improve/ final check

Project Status & Plan

PERT Chart


12/9
1/1
2/1
3/1
4/1
5/1
5/5
Fabrication
System Verification
Critical Design Review (CDR)
Complete Programs and Mount Prototyping for Sensors & Cameras
Test Motors and Propulsion Components Individually
Mount Sensors and Camera onto Drone
Integrate Motors, Prop, and remaining Electronics into Drone Hull
Test & Verify CDE 2
Test Propulsion System
Test Swarm Capabilities
Senior Design Day
1/27 Technical Data Package 1
2/24 Integrated Status Review
3/3 Technical Data Package 2
4/14 Final Acceptance Review

Critical Design Element 1

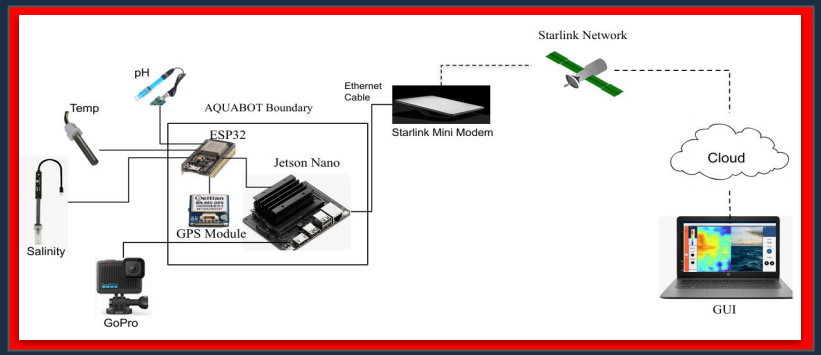
Critical Design Element (CDE-1)

CDE1 demonstrates reliable two-way data communication between AQUABOT and the ground GUI via Starlink/Wi-Fi, enabling both real-time data transmission and remote command.

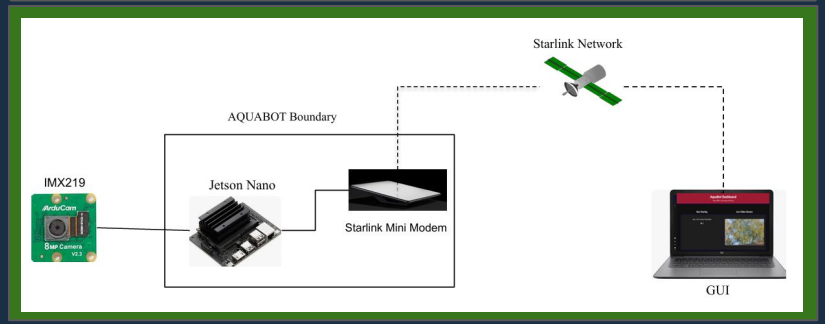
✅ **CDE1 Verified**: Demonstrated reliable two-way data communication via Starlink uplink and GUI dashboard.

- Replaced GoPro Hero 7 Black with IMX219
  - Lower power
  - Lower latency
  - CSI
- Removed Geo Overlay, will be integrated later due to time constraints
  - Didn't affect CDE1

PDR



CDR



Expected Video Bandwidth for AquaBot Live Stream (1080p H.264 WebRTC)

Raw Video (Uncompressed) Bandwidth

We start with the raw camera output:

Raw bitrate = 1920x1080×30fps×24 bits = 1,492,992,000 bits/sec ≈

1.49 Gbps uncompressed

Camera Resolution & Encoding

- **Sensor:** IMX219 (Raspberry Pi v2 camera)
- **Resolution:** 1920×1080 @ 30 FPS
- **Codec:** H.264 (NVIDIA hardware encoder)
- **Streaming Protocol:** WebRTC (real-time, low-latency)

Expected bandwidth for AquaBot

For real testing in the ocean:

Scenario	Expected
Calm Ocean	2.3 Mbps
Moderate waves	3.5 Mbps
Rapid motion	5.8 Mbps

Plus 0.25 Mbps for control + telemetry

Max Expected Bandwidth ≈ 6 Mbps

Equation used to Estimate bandwidth for 1080p H.264 WebRTC :

$$\text{Bitrate} = \text{bpp} \cdot \text{Resolution(W)} \cdot \text{Resolution(H)} \cdot \text{fps}$$

$$\text{Target Bitrate} = \text{Bitrate} \cdot (1 + \text{Overhead}) \cdot \alpha$$

******Overhead*** *= protocol + header overhead (≈ 0.10)*

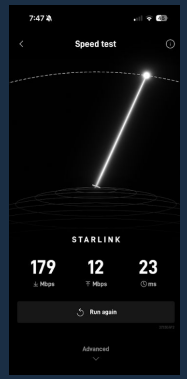
******α*** *= safety margin (≈ 0.85 for adaptive streaming)*

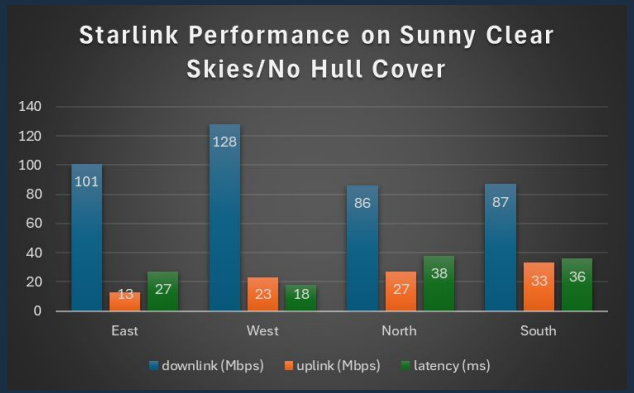
Based on validated H.264 encoding practice for 1080p30 video across varying motion complexity:

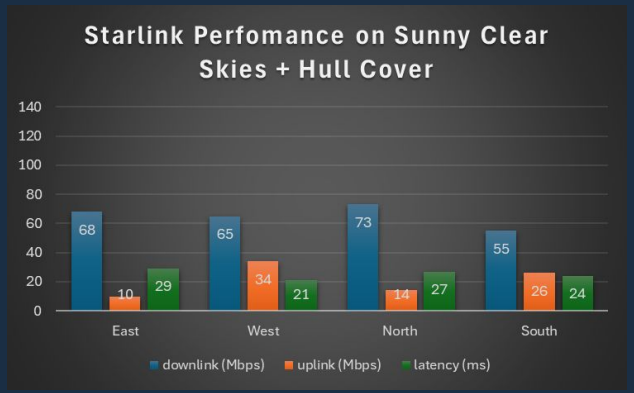
- **Calm ocean (low motion): bpp ∈ [0.03, 0.05]**
- **Moderate waves (medium motion): bpp ∈ [0.05, 0.08]**
- **Rapid motion (high motion): bpp ∈ [0.08, 0.12]**

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Starlink Analysis







- Speed test from Starlink App
- We tested in a neighborhood with lots of Wi-Fi and signals, so it's a worst-case scenario.
- Out on the ocean there are no houses or obstacles, so interference is much lower.
- Performance in deployment will be stronger than what we measured here.
- Maritime Service - Open Ocean/Stronger signal and consistent throughput.

Worst case scenario:

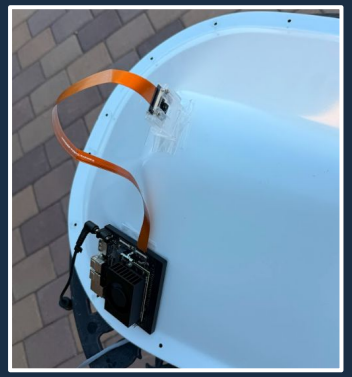
Uplink:10 Mbps
Downlink: 55 Mbps

6 Mbps < 10 Mbps

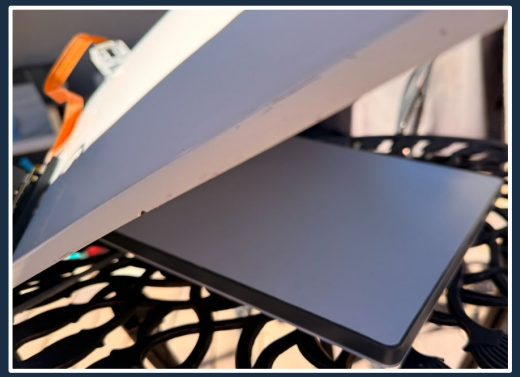
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Verification Setup: Starlink + GUI Streaming Test

1. Jetson auto boots when connected to power



2. Place Starlink under hull top

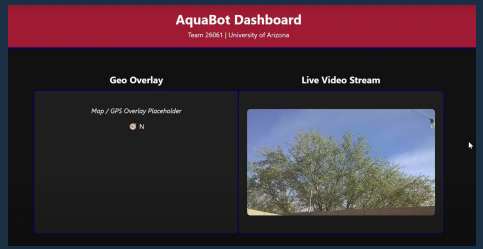


3. Run signaling_server.py & app.py on terminal

```
PS C:\Users\mayra\OneDrive - University of Arizona\Documents\AquaBotGUI
_WebRTC> python signaling_server.py
🌐 Signaling server running on ws://0.0.0.0:8765
🔵 Client connected
```

```
(venv) PS C:\Users\mayra\OneDrive - Universit
y of Arizona\Documents\AquaBotGUI_WebRTC> pyt
hon app.py
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not
 use it in a production deployment. Use a pro
duction WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.1.187:5000
```

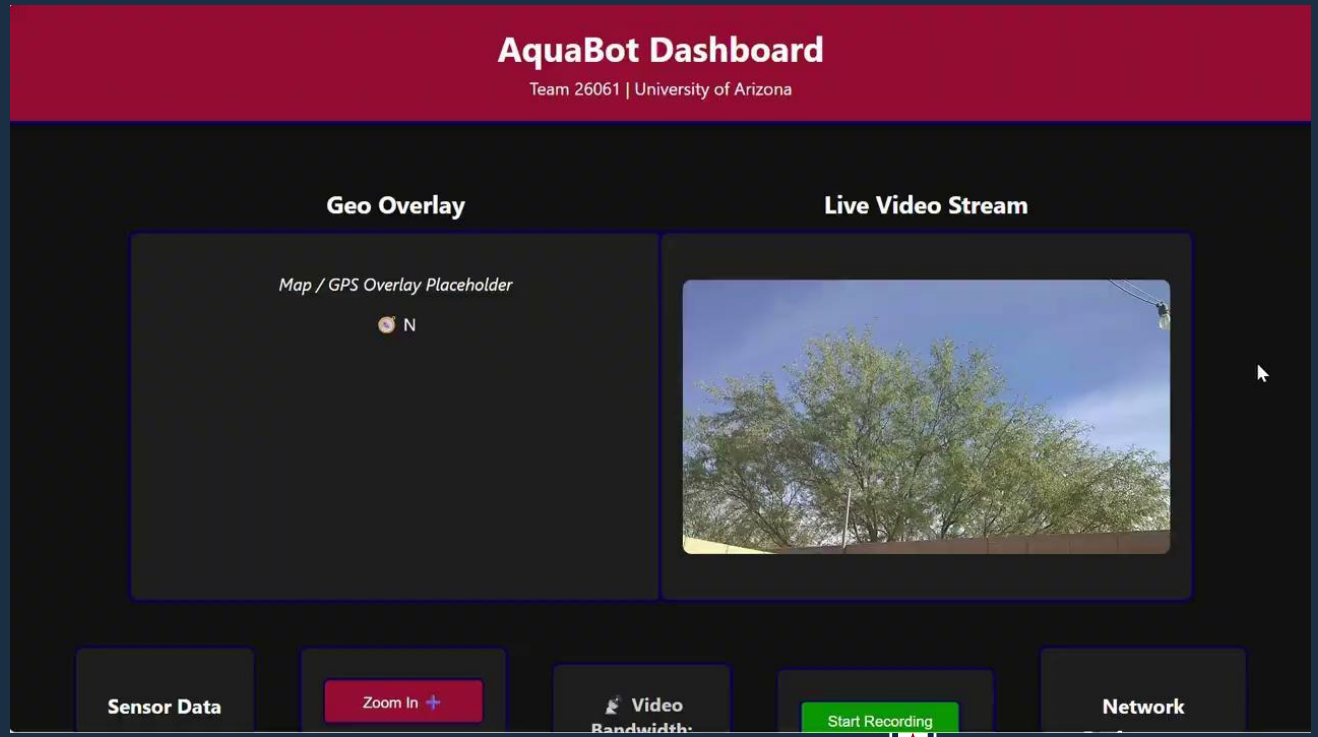
5. Go to http://192.168.1.189:5000



6. Verification checklist :
   - Recording triggered and saved locally on Jetson
   - Zoom commands received and executed
   - Run Iperf Test
   - GUI commands updated in real time

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Live GUI Demo



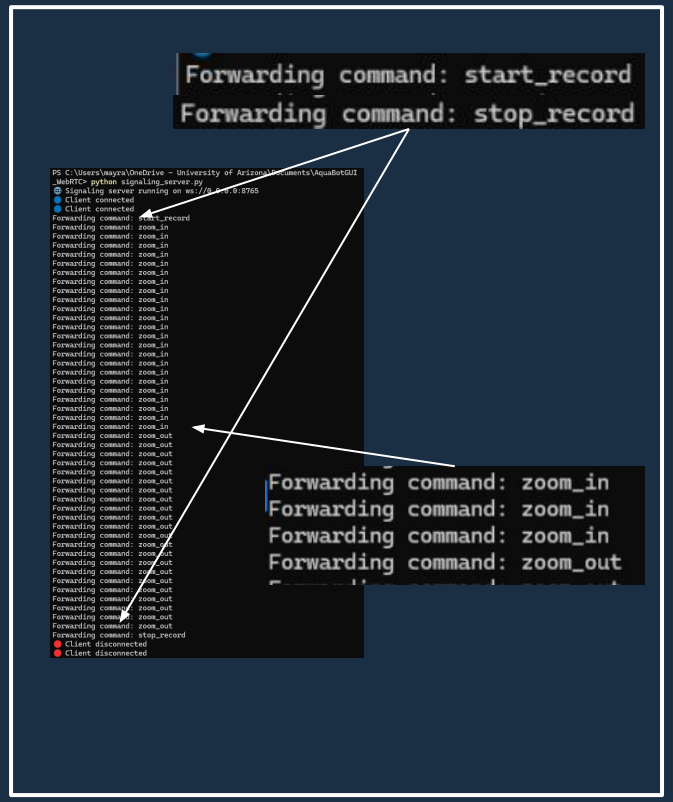
Dashboard Specs:

- Video Bandwidth and video latency
- Iperf / Network Testing
- Start recording/ stop recording
- Half geo overlay (placeholder) / half video streaming.
- Sensor data

Test conducted Nov 30, 2025 — Starlink + Jetson Orin Nano + Flask GUI

Verification

On GUI/ Laptop



Demonstrates:

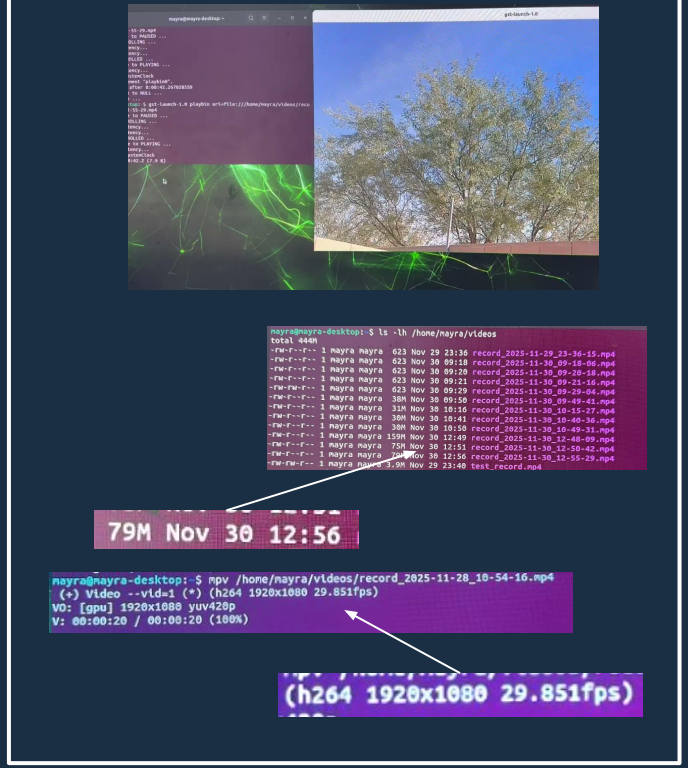
- Two-way communication
- Recording
- Saved Video files at 1080p @ 30fps

From GUI:
Video Bandwidth:
1-2 Mbps

Video Latency:
>20 ms

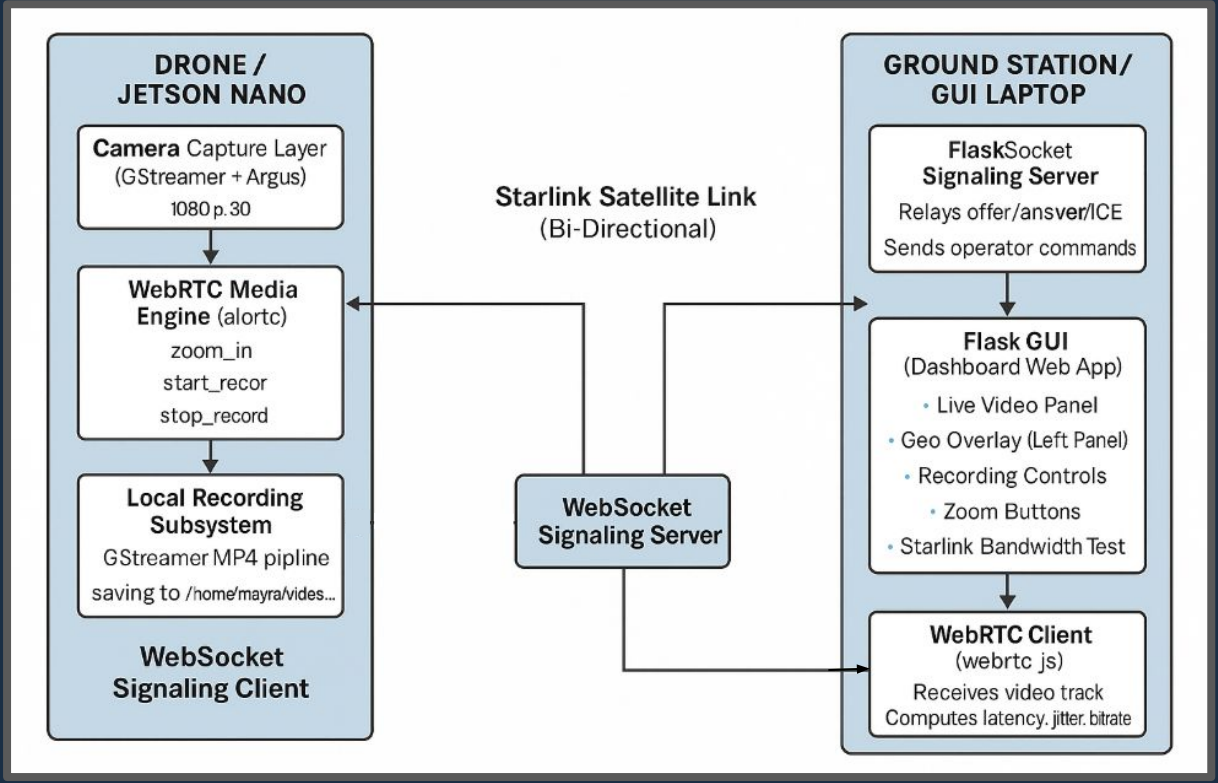
Jeston Orin Nano:
Storage 256GB

On Jetson Orin nano Monitor



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GUI Software Architecture



Backend Scripts:

- webrtc_jetson.py - Camera capture + Media Engine + Signaling
  - record_start.sh/ record_stop.sh - Local video recording
- autostart.sh - Launches Jetson service automatically on boot

Frontend Scripts:

- signaling_server.py - Initial handshake + commands
- app.py - Flask server + display logic
  - Index.html -GUI layout
- webrtc.js - WebRTC Client (video stream + operator commands)

CDE 1 Results

5.4.4 Satellite Uplink: Each AQUABOT drone shall independently transmit live and stored data to a remote ground station via its onboard Starlink.

5.1.6 Data Collection: Each AQUABOT drone shall measure temperature, salinity, pH, and capture visual imagery at a minimum of 1080p resolution, 30 fps.

- **Verified: IMX219 camera streaming at 1080p/30fps**

5.4.5 Display: The display shall show half the display with Geo-overlay and the other half will display live-streaming.

- **Verified: GUI layout splits display as required/ half live video streaming.**

5.4.1 Autonomy & Control: Each AQUABOT drone shall support both autonomous navigation and remote operator control via a satellite-based waypoint system.

- **Verified: Remote control via GUI confirmed.**

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Verified Partially Verified

CDE1: Risk and Complexity + Next Steps

CDE-1 validated reliable two-way communication on Starlink; next phases extend to multi-network.

System Complexities:

- Two-network WebRTC
- ICE/NAT traversal
- Starlink routing fluctuations
- Added data flows (video + telemetry + commands)

Risk:

Risk	Cause	Impact	Mitigation
WebRTC fails across two networks	NAT/ICE candidate rejection	Loss of remote command/video	Add TURN/STUN server, test different relay configs
Starlink connection interruptions	Satellite acquisition / weather	Increased latency, packet loss	Add buffering, dynamic bitrate (WebRTC)
Bandwidth variability	Starlink uplink fluctuations	Lower video quality	Adaptive bitrate + video fallback
Sensor + video concurrency	Multiple data channels	CPU/Network load	Prioritize RTP stream, stagger telemetry

*High Risk

Next Step:

- TURN server addition
- Remote testing
- Geo-overlay
- Full sensor integration
- CDE-2 prep

Critical Design Element 2

Critical Design Element (CDE-2)

Builds from CDE-1 two-way communication:

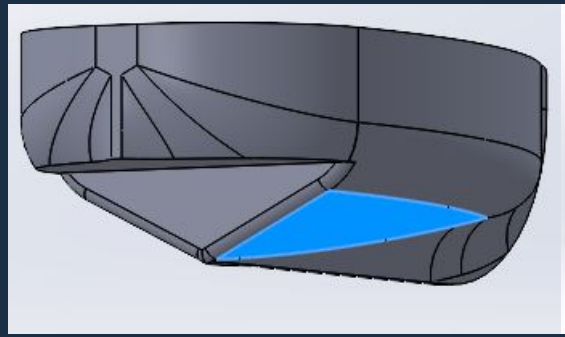
- Integrates system into hull with waterproofing
- Verifies system functionality within hull
- Transmits all sensor data in addition to camera live stream

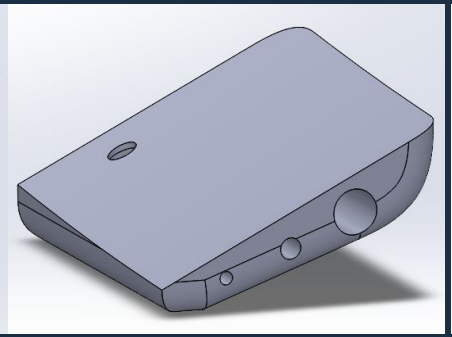
Key Steps:

- Finalize mount prototypes
- Waterproofing of camera and sensor electronics
- Safely connect and organize electronics within drones (ESP32, Jetson Nano, Starlink)
- Programming of GUI

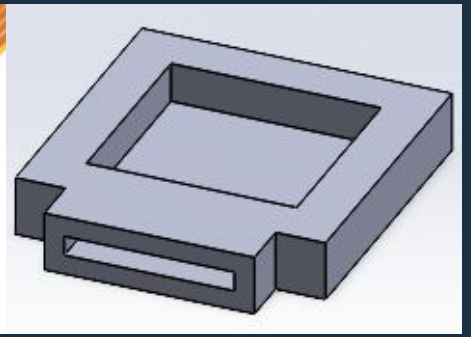
Objective:
CDE-2 will demonstrate sensor and camera connectivity to the ground station GUI, while being integrated into the drone hull itself.

Sensor and Camera Mounts









Sensor Mount:

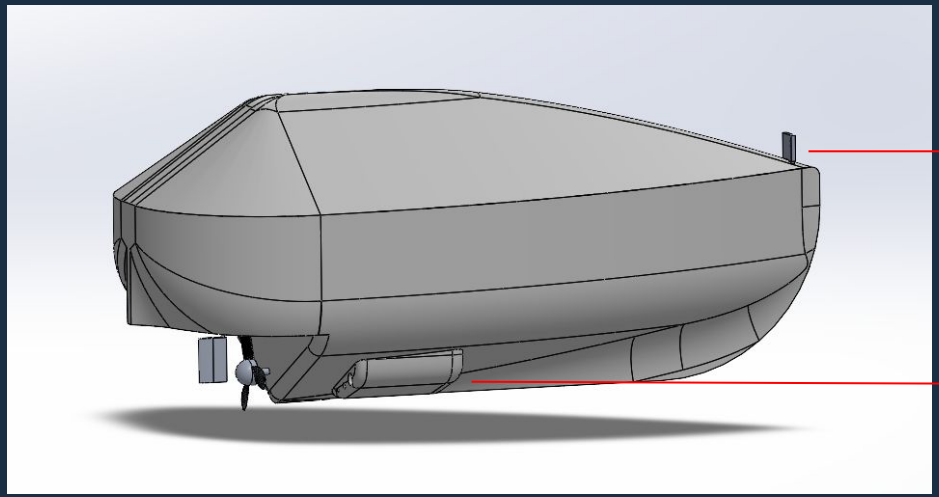
- External mount with hole for wires to connect to internal ESP32 in hull
- Recesses to fit temperature, salinity, and pH sensors, with space for wire storage in back.
- Arch can be modified to fit onto RC boats
- 3D printed with silicone coating and o-rings
- Room for additional sensors

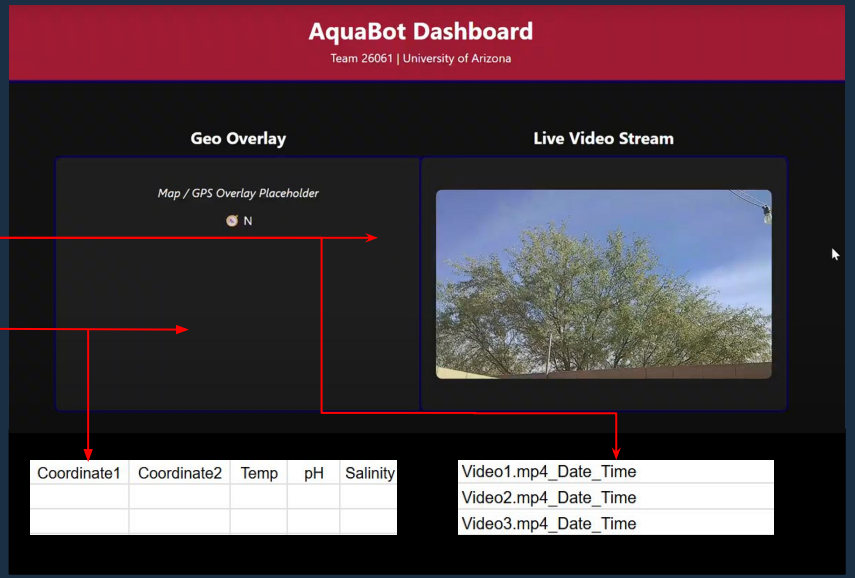
IMX219z Mount:

- Holds camera in place above drone
- Slit allows for wiring to occur within the drone.
- 3D printed with silicone coating and transparent material to protect camera

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Critical Design Element (CDE-2)





Single camera mount above, sensors mounted below

Critical Design Element (CDE-2) Verification

Procedure:

1. Place hull into pool of water of known temperature, salinity, and pH. Ensure sensors, camera, ESP32, Jetson Nano, and Starlink are all powered on and connected to ground station.
2. From ground station, ensure that sensor data is updating every 10 seconds, and camera is streaming at 30 fps and 1080p.
3. Compare sensor data to known data values for water.

Risks:

- Waterproofing
- Sensitivity, longevity of sensor run time
- Heat from drone affecting sensor values

Components needed:

- Temp, pH, and salinity sensors
- IMX219 Camera
- ESP32
- Jetson Orin Nano
- Starlink
- Ground station monitor and keyboard
- Hull
- Mounts and mounting materials

Requirements Verification

- **5.1.6 Data Collection:** Each AQUABOT drone shall measure temperature, salinity, pH, and capture visual imagery at a minimum of 1080p resolution, 30 fps.
  - Sensor + camera integration
- **5.4.5 Display:** The display shall show half the display with Geo-overlay and the other half will display live-streaming.
  - Confirms usable internal volume for all subsystems
- **5.2.4 Payload Mounts:** The hull shall provide standardized, sealed mounts for Starlink terminal, battery, and sensor payload.
  - Shows standardized sealed mounts in practice
- **5.3.2 Waterproofing:** Each aquatic AQUABOT drone shall meet IP68 ingress protection standards IAW IEC 60529:1989 + AMD1:1999 CSV. Waterproof components shall be tested in 1.5 m depths to ensure compliance.
  - Verifies waterproofing during integration sensor/camera operation

Verify with CDE2

Cost Analysis and Deliverables

Bill of Materials

Subassembly	Part	Price ($)	Purchased
1.1 Power			
	Battery	360	
	PDB	16	
1.2 Propulsion			
	Motor	90	
	Propeller	50	*
	ESA	85	
1.3 Nav/ Control			
	Arduino	17	
	IMU	17	
	GPS	8	
1.4 Comp/ Comms			
	Jetson Nano	250	
	Starlink	500	M
	Wifi Unit	5	
1.5 Sensors			
	pH	40	
	Temperature	25	
	Salinity	160	
	Camera	20	
1.6 Structure			
	Component Supp	20	*
	Hull	850	
	O-rings	5	
Total Cost ($)		2518	

* Printed or Machined Part

M Also includes monthly subscription

Cost Analysis

Subassembly	Part	Price ($)	Purchased
1.1 Power			
	Battery	360	
	PDB	16	
1.2 Propulsion			
	Motor	90	
	Propeller	50	*
	ESA	85	
1.3 Nav/ Control			
	Arduino	17	
	IMU	17	
	GPS	8	
1.4 Comp/ Comms			
	Jetson Nano	250	
	Starlink	500	M
	Wifi Unit	5	
1.5 Sensors			
	pH	40	
	Temperature	25	
	Salinity	160	
	Camera	20	
1.6 Structure			
	Component Supp	20	*
	Hull	850	
	O-rings	5	
Total Cost ($)		2518	

* Printed or Machined Part

M Also includes monthly subscription

- Highest costs:
  - Hull ($850)
  - Starlink ($500)
  - Battery ($360 *on sale)
    - Lower range or speed
- Not every drone needs a Starlink or Jetson Nano
  - Based on BW, likely ⅓
  - Average cost to $2000 per drone

Cost Analysis

Subassembly	Part	Price ($)	Purchased
1.1 Power			
	Battery	360	
	PDB	16	
1.2 Propulsion			
	Motor	90	
	Propeller	50	*
	ESA	85	
1.3 Nav/ Control			
	Arduino	17	
	IMU	17	
	GPS	8	
1.4 Comp/ Comms			
	Jetson Nano	250	
	Starlink	500	M
	Wifi Unit	5	
1.5 Sensors			
	pH	40	
	Temperature	25	
	Salinity	160	
	Camera	20	
1.6 Structure			
	Component Supp	20	*
	Hull	850	
	O-rings	5	
Total Cost ($)		2518	

* Printed or Machined Part

M Also includes monthly subscription

- Net Cost (tax + shipping): $2700
  - Will go to one complete drone
- Reserve Cost: $1300
  - $800 for:
    - Updates to 2025 Hull
    - Testing
    - Additional Materials
  - $500 for:
    - Any unforeseen costs
      - Shipping, tariffs, etc.

Deliverables

Deliverable	Date	Description
Programmed Cameras and GUI	12/9/2025	Camera connected and linked to GUI
Programmed Sensor Suite	12/9/2025	Sensors and GPS integrated
Materials CAD	12/9/2025	CAD for all major systems complete
Mounted Cameras	1/2/2026	Equipped with both storage and streaming abilities.
Communication Module	1/22/2026	Starlink integrated with all components & GS.
Propulsion System	2/5/2026	Functional propulsion system w/ motor control
Operational Ground Station	2/19/2026	GUI and backend for live data transmission
Functional Hull Prototype	3/5/2026	Hull integrated with all components
Manual	4/14/2026	Includes documentation and user guides
Project Final Report	5/5/2026	Includes summary, system specs, and results

Summary

Summary

Key Features of Our System:

- Integrated sensor suite (pH, salinity, temperature, 1080p camera).
- Starlink-based communication for live video and data transmission.
- Ground station with sensor data and camera livestream, as well as navigational control
- Reusable, impact-resistant hull.
- Swarm deployment and short-term ocean monitoring capability.

Critical Design Element (CDE-2):

Demonstrates data accuracy and live data display while sensors and camera are integrated into the hull.

Next Steps:

Finalize prototypes, begin hull fabrication and verification of performance requirements.

Questions & Discussion



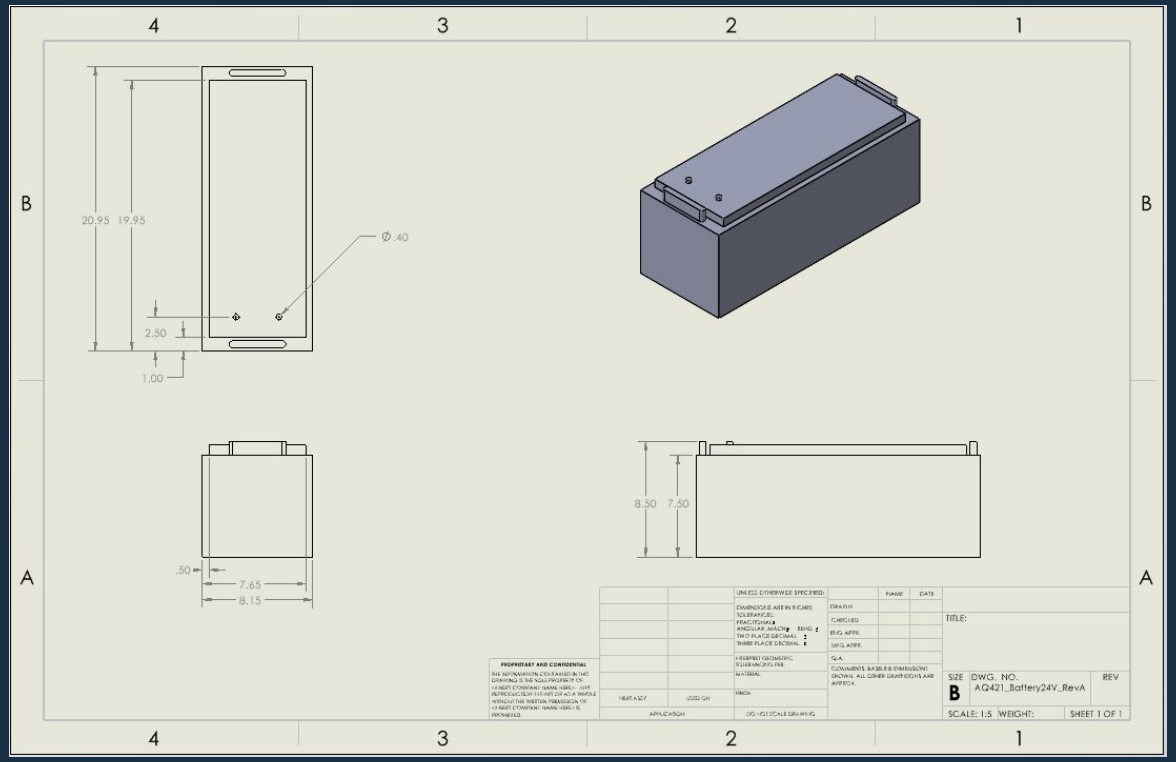
Appendix

Appendix

Part Number	Part Name	Rev Number	Status	IN WORK
1	CompletePart	1	Complete	Student1
2	WIPPart	1	WIP	
3	ObsoletePart	1	OBS	
4	To-DoPart	1	To-Do	
201	Aquabot	1	Complete	
202	ShelfAssem	1	Complete	
203	StarlinkMount	1	Complete	
401	BottomShell	1	Complete	
402	TopShell	1	Complete	
403	BottomShelf	1	Complete	
404	TopShelf	1	Complete	
405	StarlinkMountBase	1	Complete	
406	StarlinkMountBracket	1	Complete	
407	Starlink	1	Complete	
408	SupportColumns	1	Complete	
409	GPS	1	Complete	
410	IMU	1	Complete	
411	VerticalSupports	1	Complete	
412	StepDownConverter	1	Complete	
413	ShelfSupportEndCaps	1	Complete	
414	ShelfSpacer	1	Complete	
415	RaspPi	1	Complete	
416	LiPO	1	Complete	
417	JetsonNano	1	Complete	
418	MainMotor	1	Complete	
419	CameraHousing	2	Complete	
420	MotorMount	1	Complete	
421	Battery	1	Complete	
422	RCMiniPiMount	1	Complete	
501	GoPro	1	Complete	

Parts and Drawings List

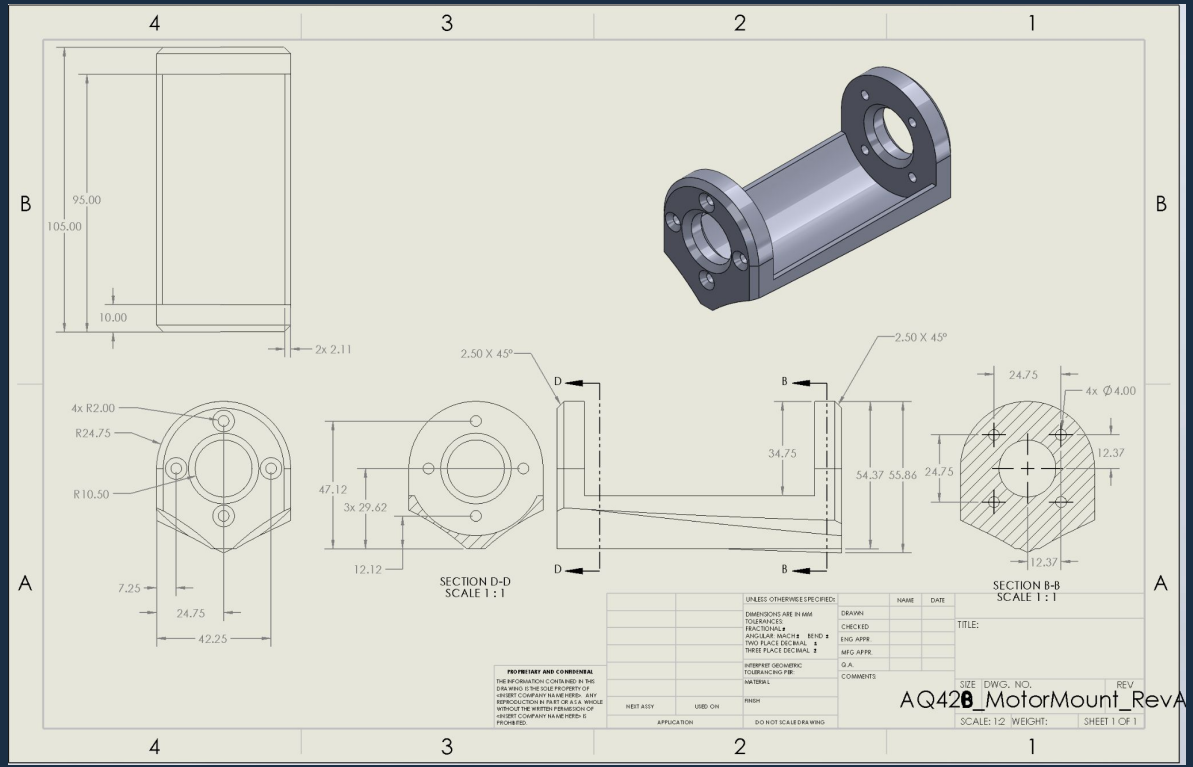
Appendix



24V Battery Drawing

THE UNIVERSITY OF ARIZONA

Appendix



Motor Mount Drawing